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New Technologies, Their Use, and Mental Health Among Adolescents: A
Meta-Analysis of the Effectiveness of Addiction Interventions

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Abstract:

Introduction: New technologies are part of daily life for most people, but there are cases of problematic use that can lead to addiction, with adolescents being the group at greatest risk. Cognitive-behavioral therapy is often cited as the most effective treatment for behavioral addictions, but there is little literature on interventions for technology addiction in adolescents.

Method: A systematic search was conducted in PsicoDoc, PsychInfo, PsychArticles, PubMed, and Medline to identify recent interventions (from 2012 to 2022) addressing addiction to new technologies (the internet, cell phones, social media, and video games) among adolescents. A random-effects meta-analysis was conducted to evaluate the effect of these interventions compared to a control group or another treatment.

Results: A total of 15 primary studies were included in the meta-analysis. A large overall effect of the treatments was found in the clinical population ($d = 0.920$, 95% CI between 0.603 and 1.238), whereas no reduction in addictive symptoms was found in preventive studies involving the general population ($d = 0.048$, 95% CI between -0.505 and 0.600). No effect was found for the potential coded moderator variables (e.g., type of treatment or type of new technology).

Discussion: There is still no consensus on the concept of technology addiction, which complicates its detection and treatment. The results of the meta-analysis should be interpreted with caution given the small sample size and the quality of the included studies. Further research is still needed to determine which interventions are most effective in adolescents.

Keywords

addiction, internet, video games, social media, cell phone, meta-analysis, adolescence

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1. Introduction

New technologies (the Internet, cell phones, social media, etc.) are an essential part of our daily lives. While it is true that the term “new technologies” may be somewhat vague and incomplete—since in a few years what is known as “new technologies” may not be the same as what we refer to today—it is a term that is informative enough to be understood by readers. Labrador & Villadangos (2009) define them as “a wide range of instruments, tools, channels, and media dedicated primarily to the use, management, presentation, communication, storage, and retrieval of information, both analog and digital” (p. 45). Another frequently used term is information and communication technologies (ICT); however, this term would exclude video games, whose ultimate purpose is nothing other than entertainment.

Labrador & Villadangos (2009) distinguish, within the context of behavioral addictions, four types of behaviors related to new technologies: the internet, cell phones, video games, and television. Currently, television use among young adolescents has declined dramatically, and the services it offers have been replaced by *online* platforms such as *Netflix* or *YouTube* (which is considered a social network). Other taxonomies, such as those by Matalí Costa (2014) or van Rooij et al. (2017), replace television with social media. This latter classification will be used in the present study. While it is true that there are other types of behavioral addictions related to new technologies, such as pornography, gambling, or compulsive shopping, these do not necessarily require this medium to occur and will therefore not be addressed in this study.

1.1. A Brief Historical Overview of New Technologies

The origins of new technologies can be traced back to the emergence of electromechanical computers, which date back to 1944 with the development of the *MARK I* computer. This machine, as is often the case with new technologies, quickly became obsolete. In 1965, the first minicomputers began to be successfully marketed, but it wasn't until 1976 that the first personal computer with an architecture similar to what we know today—the *Apple I*—appeared. From that point on, these machines underwent continuous development, leading to the powerful computers we have today, which come in both desktop and portable formats (laptops, *tablets*, and *smartphones*).

Parallel to the development of computers as advanced computing machines, the internet emerged as an advanced communication system linking computers. The origins of the internet date back to 1958 with the *Advanced Research Projects Agency* (ARPA) and the first direct connection between 23 computers across the United States via ARPANET in 1971. From the launch of ARPANET until the emergence of the *World Wide Web* (WWW) in 1991, various communication systems were developed around the world. However, it was the WWW that established a stable network connection system that gradually opened up to the public through web browsers such as those used today (e.g., *Google Chrome*, *Opera*).

Today, the internet offers a wide range of features and services, ranging from information storage—such as digital encyclopedias like *Wikipedia*—to e-commerce services for both large and small retailers. Among all these services, the internet's ability to facilitate rapid communication between people stands out in particular. Within communication services (e.g., email), what are known as social networks are currently booming.

Social networks are *software* applications—generally designed for mobile devices—that allow users to create a partially or fully public profile and establish a network of contacts with whom they can interact, communicate, and share information (Boyd & Ellison, 2007). Their origins date back to 1997 with the launch of *SixDegrees.com*, but it was in 2004, with the launch of *Facebook*, that social media platforms spread and began to be used by a large portion of the population. Some of the most influential social media platforms—which remain so today—include *YouTube* (2005), *Twitter* (2006), *Instagram* (2010), and *TikTok* (2016).

Finally, the origins of video games are unclear, but “*Noughts and Crosses*” (1952) is often cited as the first video game in history. It was a simple game of tic-tac-toe in which the player competed against the computer (*Player versus Computer, PvC*). Shortly thereafter, in 1958, the first multiplayer video game (*Player versus Player, PvP*), **Tennis for Two**, appeared. The development of video games has been closely linked to the development of computers, and it was in 1972 that the first home video game console, the *Magnavox Odyssey*, was launched. Video games as a form of leisure became popular starting that year, when the first arcade machine, *Pong*, appeared, transforming this form of entertainment into a social activity, as young people gathered at arcades. Video games and consoles continued to evolve and improve, incorporating features such as online multiplayer, to the point where professional video game competitions—or *esports*—were established.

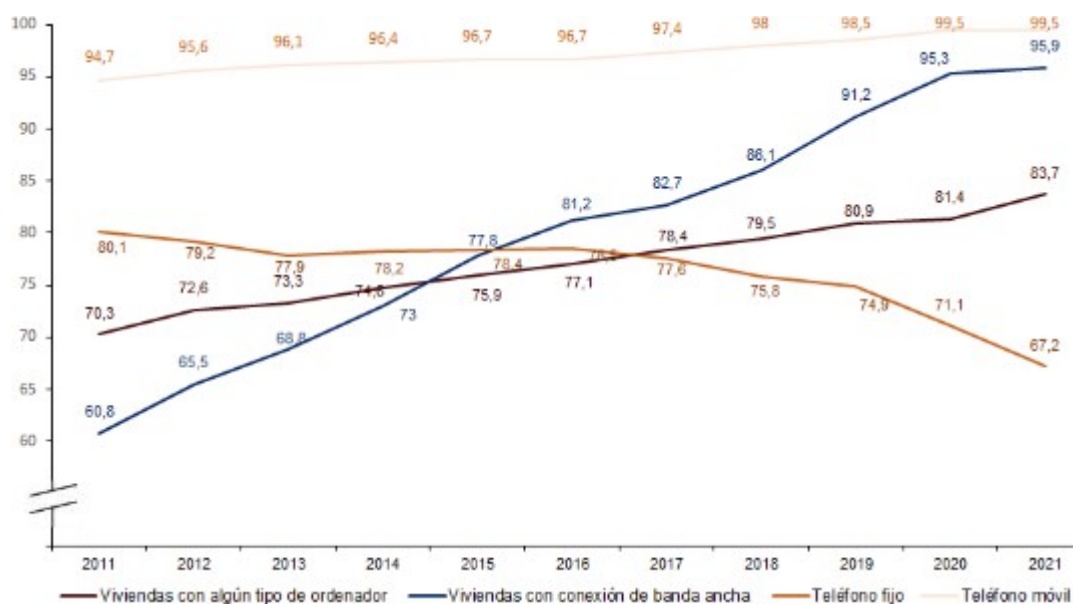
1.1.1. New Technologies Today

To assess the status of new technologies among the Spanish population, the National Institute of Statistics publishes an annual survey that tracks the use of these new technologies by the country’s population (National Institute of Statistics, 2021). As shown in Figure 1, new technologies continue to boom. 99.5% of

Spanish households have at least one cell phone, 83.7% of households have at least one computer, and 95.9% have a broadband internet connection.

Figure 1.

New Technology Ownership in Spanish Households



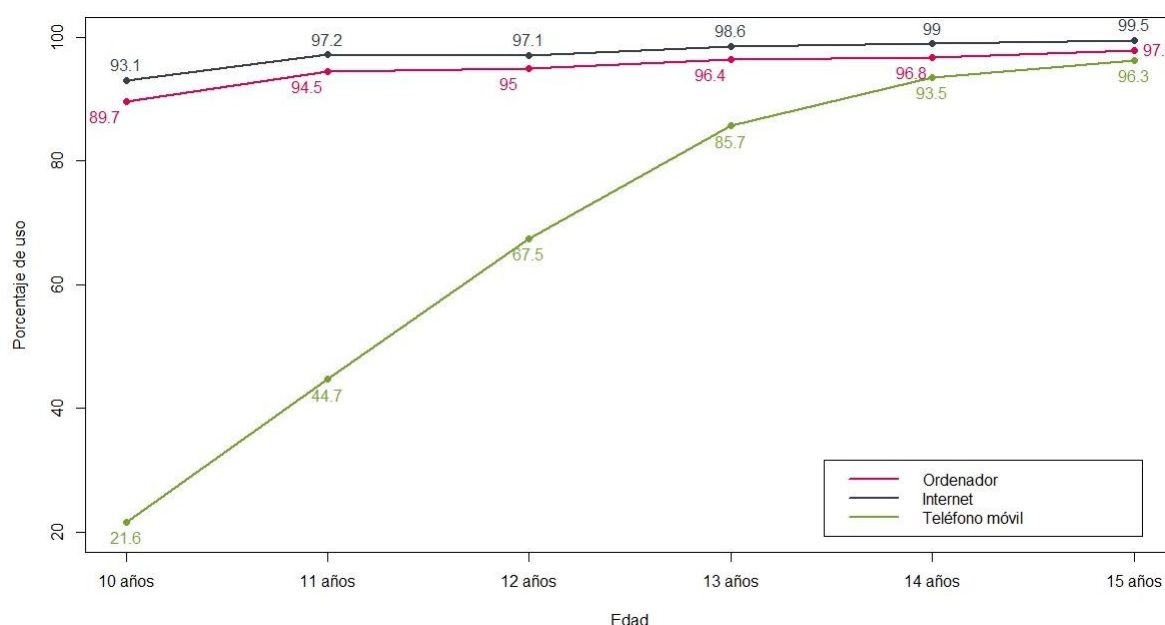
Note. Source: Survey on the Availability and Use of Information and Communication Technologies in Households (National Institute of Statistics, 2021)

Regarding the use of these new technologies, the survey reports that 93.9% of Spaniards between the ages of 16 and 74 have used a cell phone in the last three months, with 85.8% of the population using it daily. The most commonly used services are instant messaging, particularly the social media platform *WhatsApp* (90.2%), and sending or receiving emails (79.1%). Of particular note is the steady increase in e-commerce use (55.2% in 2021), with 10.5% of users having spent money on video games. In addition, 64.7% of the population has participated on a social media platform, such as *Facebook* or *Instagram*, in the last three months, with students being the group that uses them the most (96.4%).

The National Institute of Statistics does not directly collect data on the adolescent population, but it does collect annual data for the population aged 10 to 15 (Figure 2). Overall, 95.1% of children in this age group use a computer, and 97.5% use the internet. On the other hand, only 68.7% of minors use a cell phone. However, as can be seen in Figure 2, there is a rapid increase in the use of this device year after year, reaching levels comparable to those of the adult population by age 14.

Figure 2.

Use of New Technologies by Youth Aged 10 to 15



Note. Source: Author's own analysis based on data from the National Institute of Statistics (2021)

Data provided by the Telefónica Foundation (2021) offer a more precise picture of internet use among adolescents. According to this report, 78.7% of Spanish youth between the ages of 14 and 19 listened to music, programs, or podcasts online, and 83.1% also used the internet to watch multimedia content via *streaming* platforms such as *Netflix* or *Amazon Prime Video*. It also notes that 57.6%

Young people in this age group spent part of their free time playing *online* games. The following section describes the main characteristics and current status of each specific type of new technology considered in this study.

Cell phones. Cell phones have evolved from rudimentary devices whose sole function was to make and receive voice calls or, at most, send short text messages or SMS. Today, cell phones are microcomputers with direct internet access, maps, alarms, weather forecasts, stopwatches, calculators, and even high-end cameras.

Consequently, the amount of time a user can spend on their mobile phone has increased exponentially (Association for Media Research, 2022). Particularly noteworthy is the level of customization these devices offer through the download of *software* applications (apps) from the official app stores of their respective operating systems. The wide variety of apps available means that each user can use their mobile device in a more or less unique way. For many professionals, the mobile phone may be their primary work tool, while for many others, it may be their most important means of leisure or communication. In short, mobile phone use enables a very different experience among its users.

Social media. Social media platforms function as predictable and, in principle, safe community spaces, similar to traditional social media (Griffiths et al., 2014). As mentioned earlier, a considerable proportion of young Spaniards participate in one or more social media platforms. The Spanish population aged 16 to 74 spends an average of one hour and 53 minutes using social media (We Are Social & Hootsuite, 2022),

with the most widely used platforms being *WhatsApp* (91.0%), *Facebook* (73.3%), *Instagram* (71.7%), *Twitter* (46.7%), and *TikTok* (38.6%).

54.7% of Spanish adolescents aged 13 to 17 spend more than half of their time online on social media (del Barrio Fernández & Ruiz Fernández, 2014), of whom 28.68% report that social media constitutes virtually all of their online activity. The main purpose for which this demographic uses social media is to connect with friends (96.0%), although meeting new people (15.1%), “flirting” with others (14.5%), and searching for information about other people (5.7%) are also notable (del Barrio Fernández & Ruiz Fernández, 2014).

Video Games. In Spain, 42.1% of young people between the ages of 15 and 19 regularly play video games (Calderón Gómez & Gómez Miguel, 2022); specifically, 33% play video games daily. There is a significant gender difference in video game consumption, as 60.3% of men play compared to 36.3% of women. However, video game consumption is not limited solely to the act of playing; it also includes the consumption of digital products related to video games (Calderón Gómez & Gómez Miguel, 2022). 48.7% of young people in the aforementioned age group report consuming video game content through social media platforms such as *YouTube* or *Twitch*.

Video games, as a new technology intended primarily for leisure, present greater complexity. On March 22, 2022, the Culture Committee of the Congress of Deputies ratified Bill 121/000085, which amends Law 23/2011 of July 29 on legal deposit (2022). This bill classified video games as cultural heritage, which has multiple implications for the industry, such as the

the archiving of all video games produced in Spain at the National Library, or the provision of additional public funding for their creation and distribution.

This bill includes video games primarily because of titles in which narrative plays a prominent role, similar to literature or film. These are games that tell a story and in which gameplay—the defining element of this cultural heritage—plays a varying degree of importance. There are many types of video games, ranging from what are known as *arcade games* (e.g., *Tetris*)—where there is no narrative and gameplay takes center stage—to *walking simulators* (e.g., *MIND: Path to Thalamus*), where gameplay is essentially reduced to “walking forward” and it is the narrative that draws the player in.

Online multiplayer video games (or simply “multiplayer”) deserve special mention. These games emerged alongside the expansion of the internet in the 1990s with the release of *Doom* and have since become the most widely played type of game today. This type of game involves cooperation or, primarily, competition among several players connected remotely at the same time. Some of the most popular games in Spain today are *League of Legends*, *Fortnite*, *Valorant*, *Call of Duty*, and *FIFA*. The popularity of some of these competitive titles is so great that professional competitive circuits have been established, known as *esports* (electronic sports). The *esports* industry has generated over 1,000 million dollars in revenue, with a total of 532 million global viewers (Newzoo, 2022).

Also worthy of special mention is the shift in the business model that the video game industry has undergone. Previously, video game users would purchase a physical copy or a digital license and could enjoy the game’s full content. Later, what is known as *DLC* (*Downloadable Content*) began to appear,

content expansions released after the game's launch that had to be purchased separately. Today, one of the most popular business models is "*free-to-play*" (multiplayer video games—usually competitive—where players do not have to pay for the basic gaming experience). Revenue from these types of games comes from micro-payments or micro-transactions, which involve purchasing cosmetic or non-essential in-game items by paying relatively small amounts of money. Many of these games feature what are known as "loot boxes"—paid content that grants a random reward—which have been heavily criticized for their similarity to *gambling* (Close & Lloyd, 2021; Forbruker Rådet, 2022).

1.2. When New Technologies Become a Problem

Although new technologies are an essential part of daily life for many people, they are not without their problems and critics. For example, there is frequent discussion of the lack of privacy on the internet and the massive amount of data users share simply by browsing the web (Ríos Insua & Gímenez-Ullate Oteiza, 2019). In relation to mental health, the potential negative impact of these media can be divided into the macrosocial level and the personal level.

1.2.1. At the macrosocial level: Media and political perspectives

The media impact of new technologies, especially video games, often carries a negative connotation (Estallo, 2009). One of the clearest examples is news articles highlighting the consumption of violent video games by adolescents who have committed homicides. One of the first international cases in which video games were highlighted as a causal factor in a homicide was the so-called "Columbine Massacre" (Mears et al., 2017), perpetrated in 1999 by Eric D. Harris and Dylan

B. Klebold. The accusations against this form of entertainment over the years became so widespread that they led the U.S. presidency to convene a series of meetings to decide whether or not to restrict video games among the American population (Romm, 2018).

In Spain, one of the first cases in which video games were cited as a causal factor in homicides—a view still held by some media outlets today (Cerbán, 2021)—was the parricide known as the “Katana Murder,” carried out by José Rabadán Pardo in 2000. More recent cases include the homophobic mass murder known as the “Samuel Case” (Pontevedra, 2021) and the parricide committed in Elche in February 2022 (Marrahí, 2022). The media coverage of video games by these outlets appears to contradict what scientific and forensic studies suggest (Ferguson, 2008; Fox & DeLateur, 2014; Mears et al., 2017; Vuorre et al., 2022). Vuorre et al. (2022) suggest that there is no link between video game use and well-being, emphasizing that this new technology does not appear to be harmful to mental health unless there is problematic use indicating a possible addiction.

Media coverage and existing scientific evidence regarding the impact that video games may have on users’ health have led to the establishment of laws and social policies aimed at regulating and restricting the use of these technologies, especially among children and adolescents. In 2011, South Korea passed the controversial “Cinderella Law,” which prohibited minors from playing *online* games between midnight and 6:00 a.m. (Lee, 2011). However, following considerable criticism, the law was finally repealed in 2021 (Eun-byel, 2021). Much stricter is the law passed in China, which allows minors to play for just over an hour on Fridays, Saturdays, and Sundays (Hernández & Zhang, 2019).

In Europe, laws are not so much aimed at restricting usage as they are at restricting aspects related to it. In 2018, Belgium became the first country to ban the loot boxes mentioned in section 1.1, as they were considered a form of gambling (Gerken, 2018). Similarly, Spain is planning to regulate access to and the purchase of loot boxes (Ministry of Consumer Affairs, 2022). With regard to other technologies, media and scientific attention is much more limited, and consequently, there is far less legislation. In the realm of social media, Norway enacted a law in 2021 requiring content creators to disclose when a photo has been retouched for aesthetic purposes, thereby aligning with the law established in France in 2017 (Chiu, 2021).

1.2.2. On a Personal Level: Problematic Use, Addiction, and Comorbidity

The potential impact of new technologies on users' mental health has sparked interest within the scientific community. There is a wealth of literature on the problematic or pathological use of these platforms and related issues, but mental health problems have also been linked to the non-pathological use of new technologies. The British #StatusOfMind survey (Royal Society for Public Health, 2017) found an association between high levels of anxiety, depression, and *bullying* and the use of *Instagram* among adolescents. However, it also highlighted positive associations, such as an improvement in personal identity. An internal report by this social media platform concluded that *Instagram* could negatively affect adolescents' body image and, in a small number of cases, be associated with an increase in suicidal ideation (Wells et al., 2021). In general, *Instagram* could worsen the condition of adolescents who already have mental health problems. The social media platform that appears to be most closely linked to well-being is *YouTube* (Royal Society for Public Health, 2017), as it provides easier access to

reliable health-related information and helps reduce feelings of loneliness.

However, the majority of mental health problems stem from the problematic use of these technologies. Prevalence rates for online video game addiction in Europe range from 2.9% to 10% of gamers (André et al., 2020; Andrisano Ruggieri et al., 2016; González & Orgaz, 2014). With regard to social media, the prevalence of problematic use among adolescents ranges from 3.5% to 4.5% of users (Bányai et al., 2017; Luo et al., 2021). From a broader perspective, the prevalence of problematic internet use ranges from 1% to 5.5% among adolescents (Durkee et al., 2012; K. M. Kim et al., 2020; Tsitsika et al., 2014).

Addiction to video games—specifically multiplayer video games—has been the most extensively studied, to the point that it has been proposed as a distinct disorder requiring further study in the DSM-5 (American Psychiatric Association [APA], 2014): Internet Gaming Disorder, which is defined as the “persistent use of the Internet to engage in gaming, often with other players, resulting in clinically significant impairment or distress as indicated by 5 (or more) of the following criteria over a 12-month period” (APA, 2014, p. 795). The diagnostic criteria are listed in Table 1.

Table 1.

Diagnostic Criteria for Internet Gaming Disorder (APA, 2014)

-
1. Preoccupation with internet gaming. (The individual thinks about past gaming activities or anticipates playing the next game; the internet becomes the dominant activity in daily life). Note: This disorder is distinct from internet gambling, which is classified under gambling disorder.
 2. Withdrawal symptoms occur when internet gaming is removed. (These symptoms are typically described as irritability, anxiety, or sadness, but there are no physical signs of pharmacological withdrawal.
 3. Tolerance (the need to spend more and more time playing online games).
 4. Unsuccessful attempts to control internet gaming.
-

-
5. Loss of interest in previous hobbies and pastimes as a result of, and with the exception of, online gaming.
 6. Continued excessive use of online gaming despite awareness of the associated psychosocial problems associated with it.
 7. Has deceived family members, therapists, or others regarding the amount of time spent playing online games.
 8. Use of online gaming to avoid or alleviate negative emotions (e.g., feelings of helplessness, guilt, anxiety).
 9. Has jeopardized or lost a significant relationship, job, or educational or career opportunity due to participation in online gaming.

Note: Only non-gambling online games are included in this disorder. The use of the internet to perform activities required for a business or profession is not included; nor is the disorder intended to encompass other recreational or social uses of the internet. Similarly, adult websites are excluded.

Regarding the possibility of addiction to other types of new technology, or even to the internet in general, the DSM-5 is very brief. Regarding social media, it states that “it is not considered analogous to internet gaming disorder” (p. 797). However, various authors have proposed diagnostic criteria for internet addiction (Goldberg, 1995; Ko et al., 2009) and for mobile phone addiction (Muñoz-Rivas & Agustín, 2005). Tables 2 and 3 present these criteria for the internet and mobile phones, respectively.

Table 2.

Diagnostic criteria for internet addiction (Ko et al., 2009)

-
1. Excessive preoccupation with Internet activities.
 2. Difficulty controlling the urge to use the Internet.
 3. Tolerance, defined as a need to increase the amount of time spent online to experience the same satisfaction.
 4. Withdrawal syndrome, characterized by dysphoria, anxiety, irritability, and boredom that subside when Internet use resumes.
 5. Staying online longer than intended.
 6. Experiencing a persistent desire or unsuccessful attempts to stop or reduce Internet use.
-

-
7. Spending an excessive amount of time engaging in activities that involve Internet use.
 8. Excessive effort to gain access to the Internet.
 9. Continued Internet use despite awareness of the physical and psychological consequences of excessive Internet use.
-

Note: To diagnose Internet addiction, there must be a pattern of maladaptive use that causes significant distress for more than 3 months and meets 6 or more of the above criteria.

Table 3.

Diagnostic Criteria for Cell Phone Addiction (Muñoz-Rivas and Agustín, 2005)

-
- a) Tolerance: stimuli progressively lose their rewarding value and, as a result, must be intensified. In the case of mobile phone addicts, the sense of well-being produced by communication behaviors is quickly lost, so the behaviors that serve as reinforcers must be repeated with increasing frequency.
 - b) Withdrawal: when the person is unable to use the device, they experience symptoms similar to those seen in withdrawal from other substances (anxiety, nervousness, general malaise...).
 - c) Greater use than initially intended: the use of these devices can quickly become essential for the person, who ends up spending more hours on them than intended.
 - d) Desire to control or stop its use: some users try ineffective strategies such as putting their phone on silent, turning it off, or blocking incoming calls... However, when these control mechanisms are removed, the need to respond to missed calls and received messages resurfaces...
 - e) Decrease in social and recreational activities: excessive phone use can have significant repercussions on a person's social, family, work, and/or school life.
 - f) Persistence of the behavior despite awareness of its negative effects.
-

Note: To be considered an addiction, two or more of these symptoms must have been present during the past 12 months.

A number of risk factors have been associated with addiction to new technologies. In general, regarding internet addiction, being male and high mobile phone use have been reported as risk factors (S. W. Choi et al., 2015), as well as high scores on the personality dimensions of disinhibition, extraversion, neuroticism, and narcissism (Peris et al., 2020) and not having had a secure attachment figure during childhood (Karaer & Akdemir, 2019; Matalí Costa, 2014). Regarding problematic or abusive cell phone use, being female

and having a prior history of problematic internet use (S. W. Choi et al., 2015), as well as the personality traits already mentioned for internet addiction (disinhibition, neuroticism, extraversion, and narcissism) (Peris et al., 2020).

With regard to social media addiction, research has identified the following risk factors: being female, having a history of behavioral problems, hyperactivity, impulsivity, and sedentary behavior (Mérelle et al., 2017; Wu et al., 2013). The personality traits associated with this problem (Peris et al., 2020) are disinhibition, neuroticism, and extraversion. Finally, for online video game addiction, reported risk factors include being male, a history of behavioral problems, a sedentary lifestyle, low socialization, attention-deficit/hyperactivity disorder (ADHD), and parental anxiety (Mérelle et al., 2017; Wartberg et al., 2017). The personality traits associated with this problem (Peris et al., 2020) are disinhibition and neuroticism.

The high comorbidity between addiction to new technologies and other mental health problems deserves special mention (Aboujaoude, 2010). This fact has sparked significant controversy over whether addiction to new technologies is a disorder in its own right or whether it is a symptom of other disorders (Matalí Costa, 2014). Although models in which addiction is a predictor of poor mental health—rather than the reverse—have been successfully evaluated (Ciarrochi et al., 2016), there is still no consensus on this issue. Many mental health problems have been reported as comorbid with internet addiction. To name a few, associations have been found with various anxiety disorders and states (S. W. Choi et al., 2015), as well as major depressive disorder and alexithymia (Karaer & Akdemir, 2019; van Rooij et al., 2017).

Similarly, smartphone addiction has also been associated with mental health problems such as anxiety disorders, alcohol dependence, and alexithymia (S. W. Choi et al., 2015; Gündoğmuş et al., 2021). Social media addiction has been linked to mood disorders, specifically depressive states (van Rooij et al., 2017). Video game abuse has been linked to mental health problems such as suicidal ideation (Mérelle et al., 2017), depression and sleep-related problems (Severo et al., 2020), and attention-deficit disorder (Stetina et al., 2011).

1.3. Intervention for Behavioral Addictions

The inclusion of a chapter on substance-related and addictive disorders in the DSM-5 (APA, 2014) stems from the similarities found between substance-related addictive disorders and behavioral or non-substance-related addictive disorders. These similarities have been observed in various areas, such as phenomenology, tolerance, comorbid disorders, neurobiological mechanisms, and response to treatment (Grant et al., 2010).

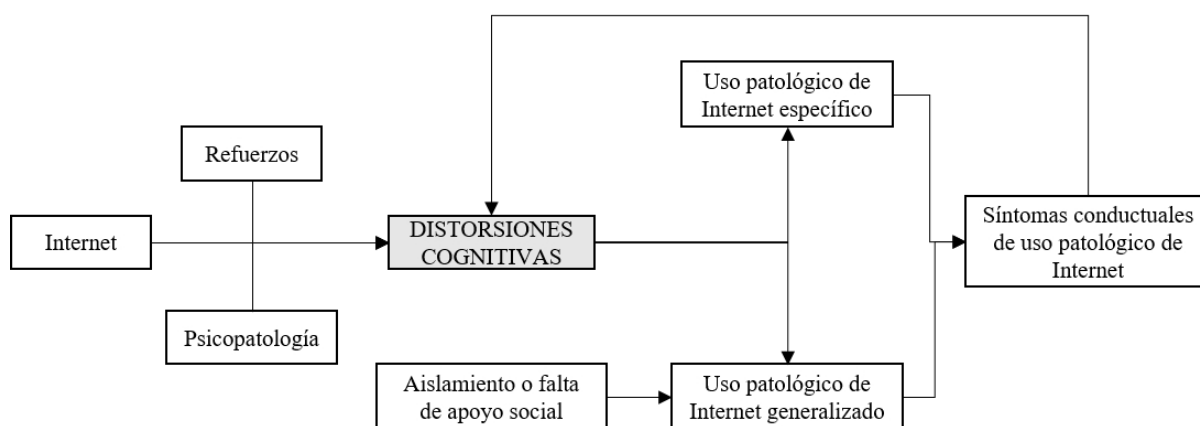
Both pharmacological (Pallesen et al., 2007) and psychological (Pallesen et al., 2005; Toneatto & Ladoceur, 2003) treatments have been found to be effective in addressing behavioral addictive disorders (e.g., gambling). Among psychological treatments, cognitive-behavioral therapy appears to be the most effective (Pallesen et al., 2005). However, depending on the theoretical model used, there are many other interventions that are more or less widely used in professional practice, such as dialectical behavior therapy, self-help or support groups, problem-solving therapy, cognitive restructuring, and relapse prevention (Ajegena et al., 2018; Bombín, 2010; Díez et al., 2015).

1.3.1. Addiction to New Technologies

Several explanatory models have been proposed for addiction to new technologies, particularly for internet addiction (in general terms) and online gaming disorder. Díaz Hurtado (2014) proposes a model that integrates the various proposals of other authors, particularly that of Davis (2001). As shown in Figure 3, easy access to the internet, psychopathological vulnerability traits, and social/environmental contingencies lead to a system of cognitive distortions that result in both specific (e.g., video games, social media) and general pathological internet use. This pathological use is also fueled by social isolation and ultimately produces behavioral symptoms that, in turn, reinforce the system of cognitive distortions. It is worth noting that the contingencies referred to in the model are the low-cost reinforcements provided by internet use, as opposed to the difficulty of receiving social and/or environmental reinforcements offline. Consistent with this model, Brand et al. (2014) highlight poor coping skills and the expectation that the internet can improve or worsen one's mood as mediating variables between vulnerability factors and addiction.

Figure 3.

Explanatory Model of Internet Addiction



Note. Source: Díaz Hurtado, R. M. (2014). Intervention with adolescents. In X. Carbonell (Ed.), *Technological addictions: What they are and how to treat them* (p. 60). Síntesis.

Although general models exist, such as the one mentioned above, specific models have also been proposed depending on the new technology in question. With regard to cell phones, social pressure arising from the use of this device is particularly noteworthy (Muñoz-Rivas et al., 2009). Especially among adolescents, owning a cell phone serves as a mechanism for social inclusion and status enhancement. Another model (Wei et al., 2017), this time concerning internet gaming disorder, suggests the existence of three neurocognitive systems involved: (1) the impulsive system, which is thought to function in a hyperactive manner; (2) the reflective system, which is thought to function hypoactively; and (3) the interoceptive attention system, which is thought to enhance the activity of the impulsive system.

The models described here, as well as others (e.g., Young & Brand, 2017), suggest that the most appropriate intervention is cognitive-behavioral therapy, as it addresses the core components of the model. Given the importance of the family context in adolescent addiction, family therapies are also recommended (Díaz Hurtado, 2014). However, there are many therapeutic approaches used, such as psychoeducation, self-help, or hospitalization (Díaz Hurtado, 2014). In Asian countries (China, Japan, South Korea), therapeutic camps or “*boot camps*” are particularly common; these may follow a regimen similar to military service (Koo et al., 2011) or take a much more therapeutic approach (Huang et al., 2022; Sakuma et al., 2017).

1.4. Objectives and Hypotheses

The objective of this study is to evaluate the combined effectiveness of recent interventions targeting addiction to new technologies (the internet, cell phones, video games, and social media) in adolescents. To this end, a meta-analysis will be conducted of interventions published in recent years (between 2012 and 2022), and various moderation analyses

moderation analyses to explain part of the heterogeneity in the results. Based on the literature on this type of intervention, cognitive-behavioral therapy is expected to be more effective than other therapies. On the other hand, given the similarity among the types of new technologies, no differences are expected in the effectiveness of interventions across different types of technology. Nor are differences expected based on sex or age.

2. Method

2.1. Study Selection

2.1.1. Search Procedure

Given the rapidly changing nature of new technologies, a systematic search of databases was conducted from 2012 through June 2022, with the aim of including only the most recent primary studies. The selected databases were *PsycInfo*, *PsycArticles*, *PsicoDoc*, *PubMed*, and *Medline*, using the following search terms: “*treatment*” AND “*addiction*” AND (“*internet*” OR “*computer games*” OR “*digital gaming*” OR “*online social networks*” OR “*social media*” OR “*smartphone use*” OR “*facebook*” OR “*instagram*” OR “*Twitter*” OR “*TikTok*”). For the PsicoDoc database, an additional search was conducted using the Spanish terms: “*tratamiento*” AND “*adicción*” AND (“*internet*” OR “*videojuegos*” OR “*redes sociales*” OR “*teléfono* móvil**” OR “*Facebook*” OR “*Instagram*” OR “*Twitter*” OR “*TikTok*”).

The terms used were extracted from the “Thesaurus of Indexed Psychological Terms” (American Psychological Association, 2022), and the search was supplemented with synonyms frequently used in the scientific literature, as well as the names of the most widely used social media platforms today. Through this procedure, 1,261 bibliographic references were identified. An additional 5 references were identified by

search for studies included in a previous meta-analysis (Malinauskas & Malinauskiene, 2019), bringing the total number of primary sources screened to 1,266.

The initial screening of articles was conducted by reviewing the title and/or abstract, selecting those that appeared to refer to an intervention for treating addiction to new technologies. Following this second selection of articles, and after removing duplicate sources, the final selection was made by reading the full texts of those articles, selecting those that met the inclusion criteria outlined in section 2.1.2. Articles for which the full text could not be accessed were excluded. Figure 4 shows a summary and the results of the process. Finally, the variables (section 2.2) and data relevant to the meta-analysis were coded.

2.1.2. Inclusion Criteria

Only publications meeting the following criteria were included: (1) the publication is peer-reviewed, as this is a basic requirement for scientific quality; (2) the text is written in Spanish or English; (3) the study sample consists of adolescents and pre-adolescents between the ages of 10 and 19; (4) the research design includes a post-intervention comparison between an experimental or treatment group and a control group or another treatment group; (5) the groups appear comparable (as assessed by a non-significant pre-treatment comparison); (6) the description of the treatment is clear enough to be categorized; (7) it contains at least one measurement instrument that assesses addiction to some type of new technology; and (8) the reports must contain the data necessary to calculate the effect size.

2.2. Moderating Variables

Moderating variables are those that explain or predict the variability among the different primary studies included in the analysis. The study characteristics considered as potential moderating variables for this review can be grouped into three categories (Lipsey & Wilson, 2001): substantive variables, methodological variables, and extrinsic variables. Substantive variables refer to the sociodemographic characteristics of the sample. In this meta-analysis, we considered the two variables that are typically included in meta-analyses: (1) gender (defined as the proportion of women in the sample) and (2) age (defined as the mean age of the sample). The analysis of these types of variables, which are presented at the participant level, is conducted using averages or means, since individual data from the primary studies are not typically available in a meta-analysis.

Methodological variables refer to the research design decisions made by the authors of the primary studies. In our meta-analysis, we coded: (3) the target population, categorized as non-clinical (primary/secondary prevention intervention) or clinical (therapeutic intervention); (4) the type of new technology whose addiction was intended to be treated, categorized as the internet, video games, mobile phones, or social media; (5) the type of control or comparison group, categorized as treated or untreated; and (6) the type of therapy applied. The last type of variables—extrinsic variables—refer to characteristics external to the sample or study design. In this study, we considered only (7) the methodological quality of the primary studies. Quality was assessed using the instrument developed by López-López et al. (2013), which evaluates threats to the validity of empirical studies on the efficacy of

pharmacological and psychological interventions for mental disorders. It takes the form of a “Yes/No” *checklist* with 10 items, with one point awarded for each item answered “Yes.”

2.3. Statistical Methods

To quantify the effect of the intervention, the standardized mean difference (d) proposed by Cohen (1988) was used as the effect size index; specifically, the correction proposed by Hedges (1981) was employed, which established an unbiased estimator of the parameter. This effect size refers to the difference between the means on the addiction measurement instruments between the experimental group and the control group on the posttest. Since most instruments indicate that a higher score corresponds to a higher level of addiction, the sign of the effect size index has been inverted to facilitate interpretation. Thus, higher values of d indicate a greater effect of the intervention on the sample. The effect sizes for the various studies were calculated using the *Campbell Collaboration's online* effect size calculator (Lipsey & Wilson, 2001).

Given the large differences in sample sizes that typically exist among the primary studies included in a meta-analysis, meta-analytic estimates are calculated by weighting the studies according to their variability. For this meta-analysis, weighting was performed using a random-effects model, which assumes that the effect sizes of the primary studies are estimators of a population of parameters, rather than of a single parameter. This model is the method of choice in most meta-analyses in psychology for two fundamental reasons that contrast with the fixed-effects model (Borenstein et al., 2010). First, this model allows for greater generalizability, and second, it is more conservative in statistical inferences.

The choice of this model implies that the weighting of the studies is based on two sources of variability: within-study variance (specific to each estimate) and between-study variance or τ^2 (common to all studies). The between-study variance was estimated using the restricted maximum likelihood method¹, which has been shown to be the most appropriate method in most cases (Viechtbauer, 2005). Thus, the weighting factor is what is known as the weight, and it is calculated as the inverse of the sampling or within-study variance plus the between-study variance.

The homogeneity of effect sizes was assessed by calculating the Q statistic (Hedges & Olkin, 1985), which follows a χ^2 distribution with $k-1$ degrees of freedom, where k is the number of effect sizes. This statistic allows for the calculation of I^2 , which indicates the percentage by which the observed variability exceeds that expected by chance. Regarding the statistical analysis of moderator variables, those with a categorical measurement level were tested using a weighted analysis of variance procedure, while those with a quantitative measurement level were tested using a meta-regression analysis.

The tests were conducted using the statistical *software R*, specifically the *metafor* package (Viechtbauer, 2010), with the exception of the tests for categorical moderators, which were performed using Wilson's (2021) macros for *IBM SPSS Statistics*. Power calculations for the moderation tests were performed following the procedure proposed by Valentine et al. (2010), with the exception of the meta-regression tests, since calculating the power for this type of test requires the original variance-covariance matrices (Valentine et al., 2010). For reasons of

¹ The values shown in the *forest plot* figures were obtained using a τ^2 estimated via the method of moments (DerSimonian & Laird, 1986), as this is the only method supported by the *software* used to generate them. Therefore, the values in these figures may differ slightly from those presented in the text.

transparency and replicability, the R code used can be found in Appendix 1 and the SPSS syntax in Appendix 2. The *forest plots* were generated using *Review Manager software*.

2.3.1. Publication Bias

Publication bias is one of the greatest threats to meta-analysis and, as a result, is one of the phenomena that has received the most attention from methodologists and researchers (Botella & Sánchez-Meca, 2015). In the process of scientific production, not all research studies result in publication. There are various reasons why a study may not be published: it may be the researcher themselves who decides not to submit their results for publication; or it may be the editorial board of the scientific journal that rejects the manuscript. It is well known within the scientific community that journals are reluctant to publish research whose results are not significant, impactful, or novel (Francis, 2012). This means that the studies ultimately included in a meta-analysis are those that are not only significant but also have larger effect sizes. This phenomenon is known as publication bias, and it potentially leads to a biased estimate of the effect size in the meta-analysis.

There are several methods for assessing and attempting to correct publication bias. In this meta-analysis, we will assess the possible presence of bias using the *safety numbers* technique and by analyzing the symmetry of the *funnel plot*. The *safety number* is the most classic technique, initially developed by Rosenthal (1979), which determines how many studies with non-significant results—that have remained unpublished—would be needed to nullify the effect of the studies included in the meta-analysis. To assess the feasibility of the *safety number*—that is, the existence of j studies that would nullify the effect—Rosenthal proposed setting a threshold of $5 \cdot j + 10$, such that if the *safety number*

exceeds the threshold value, we can assume that the meta-analysis is robust to the effect of publication bias. The *safety number* technique, particularly Rosenthal's procedure, is beginning to be considered obsolete; therefore, this technique will also be applied using the procedures proposed by Orwin (1983) and Rosenberg (2005), as a complement to the symmetry analysis of the *funnel plot*.

The *funnel plot* (Light & Pillemer, 1984) is currently the primary graphical tool available for examining the possible presence of publication bias. It consists of a scatter plot with the estimated effect size on the x-axis and the standard error of the study on the y-axis. The absence of studies on the side opposite the expected result is considered evidence of publication bias. There are additional statistical methods that aid in the interpretation of the *funnel plot*. First, Kendall's tau ordinal correlation coefficient (Begg & Mazumdar, 1994) assesses asymmetry by examining the agreement of studies on either side of the mean effect size. Second and finally, Egger's linear regression analysis (Egger et al., 1997) between effect size and precision indicates symmetry (and thus the absence of bias) if the line passes through the origin.

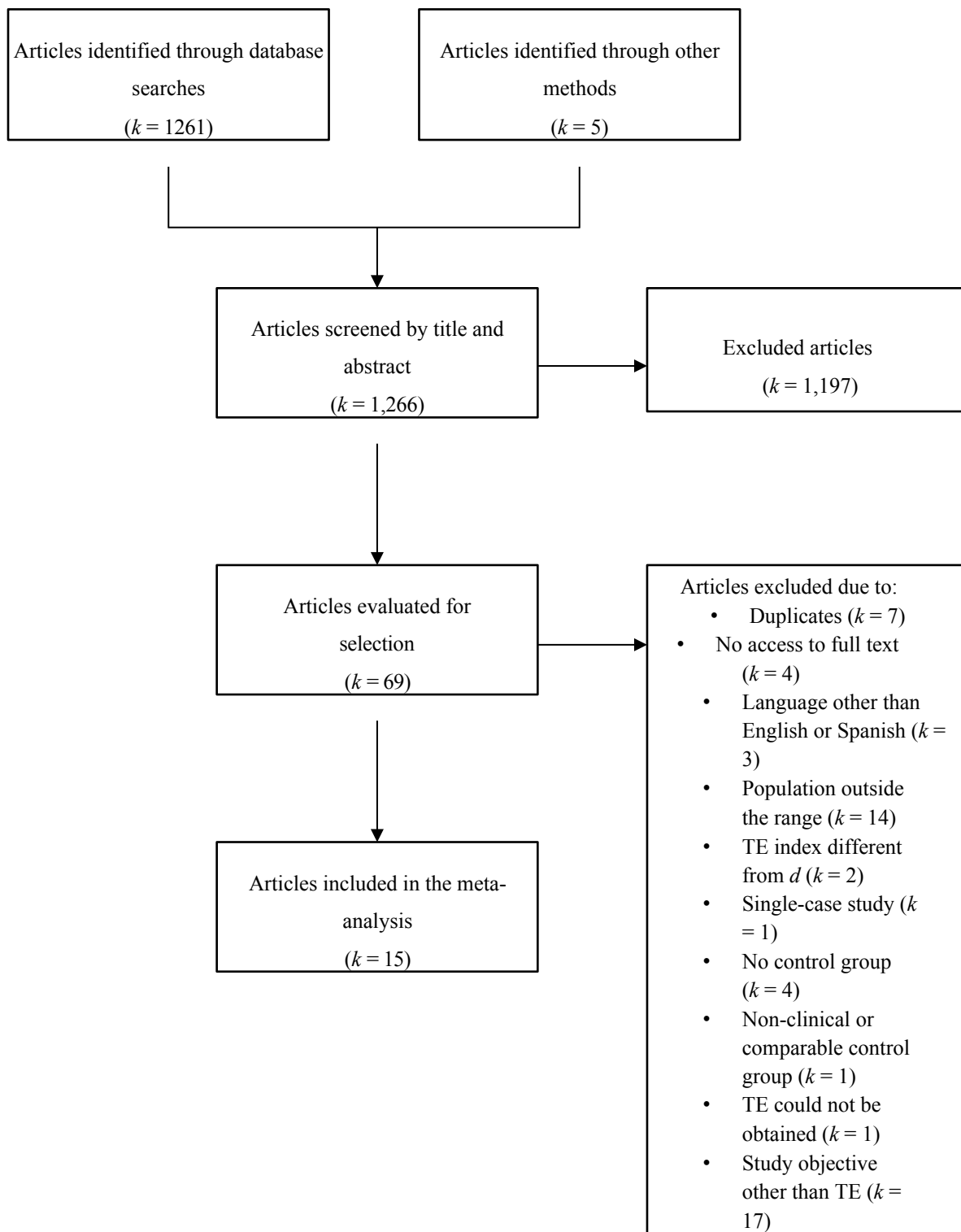
As for techniques to correct for publication bias, the most widely used is "*trim and fill*" (Duval & Tweedie, 2000a, 2000b), which is based on the *funnel plot*. The procedure consists of "trimming" the most extreme values at the tail of the effect distribution until there is no asymmetry, and then re-estimating the effect size, which is theoretically unbiased. To avoid an artificial reduction in the variance of the estimate, a "fill" process is carried out by imputing the trimmed studies but with the opposite sign. These imputed studies are used solely for the calculation of the variance.

3. Results

3.1. Description of the Selected Studies

After screening the 1,266 articles initially identified through a database search, 15 articles were selected for inclusion in the meta-analysis. The 15 articles contain a total of 17 comparisons that meet the criteria for analysis, since the study by Pornnoppadol et al. (2020) includes multiple comparisons with a single control group. Therefore, to preserve the principle of local independence—that is, that all observations included in the meta-analytic comparisons are independent—only the main comparison of the study (combination therapy) was included in the meta-analysis. Figure 4 shows a flowchart summarizing the study screening and selection process. It is worth noting the number of articles whose objectives aligned with those of this meta-analysis but which were excluded because they employed low-quality procedures and designs ($k = 6$). This group consists mainly of studies with a pre-post design without a control group.

The descriptive characteristics of the selected studies are presented in Table 4. Of particular note are the three studies involving non-clinical populations (del-Rey-Alamillo et al., 2012; Lindenberg et al., 2022; Walther et al., 2014), which do not aim to treat adolescents in whom a potential addiction problem has been detected, but rather consist of large-scale educational interventions that seek to improve habits regarding the use of new technologies and prevent potential addictions. For this reason, we have decided to exclude these studies from the other moderation analyses not related to the study's objective or population.

Figure 4.*Flowchart of study selection*

Note: Procedure adapted from the *Preferred Reporting Items for Systematic Reviews and Meta-Analyses* (PRISMA) (Moher et al., 2009)

Table 4.*Descriptive characteristics of the studies included in the meta-analysis*

Study	Year	Gender	Age	Population	NT	Treatment	Control	Measurement Tool	Country	TE	Variance	<i>n_t</i>	<i>n_c</i>	Cal.
Agbaria	2022	30.0	13.45	C	I	TCC	NT	YIAS	Israel	1,887	0.0361	80	80	6
Bong et al.	2021	50.7	12.57	C	I + TM	TCC + IA	TP	YIAS + SAPS	Korea	0.274	0.0293	67	71	5
del-Rey-Alamillo et al.	2012	45.9	13.80	N/A	I	IE	NT	CERI	Spain	0.015	0.0050	595	298	4
Hong et al.	2020	0.0	15.70	C	JL	TCC + EF	TP	YIAS	Korea	0.775	0.0860	25	25	5
S. M. Kim et al.	2012	0.0	16.05	C	JL	TCC + M	M	YIAS	Korea	0.711	0.0654	32	33	6
P. W. Kim et al.	2013	0.0	17.50	C	JL	IA	NT	Game time	Korea	0.583	0.0712	27	32	5
Li & Wang	2013	0.0	15.52	C	JL	TCC	NT	YIAS + OGCAS	China	0.393	0.1457	14	14	6
Lindeberg et al.	2022	54.3	15.11	NC	I + JL	IE	NT	CSAS	Germany	0.117	0.0099	167	255	7
Liu et al.	2015	17.4	15.35	C	I	TF	NT	APIUS	China	2,401	0.1503	21	25	3
Nielsen et al.	2021	NR	14.9	C	JL	TF	TP	Petry's IGD Scale	Switzerland	0.467	0.1247	12	26	5
Pornnoppadol et al.	2020	7.7	14.33	C	JL	CT + EP	NT	GAST	Thailand	1,005	0.0808	26	30	7
Shin & Jang	2016	46.9	14.81	C	TM	TPD	NT	S-Scale	Korea	0.771	0.1343	16	16	4
Torres-Rodríguez et al.	2018	0.0	14.96	C	JL	TCC	TP	IGD-20	Spain	1,527	0.1668	16	15	3
Uysal & Balci	2018	29.8	13.07	C	I	IE	NT	IAS	Turkey	0.514	0.0492	41	43	6
Walther et al.	2014	50.5	12.00	NC	I + JL	IE	NT	ISS + KFN-CSAS-II	Germany	0.010	0.0220	804	1039	6

Note: Gender is quantified as the percentage of women in the sample; NR = Not Reported. Age is quantified as the mean age of the sample. Population: C = Clinical / Subclinical; NC = Non-Clinical. NT (Addiction to which New Technology was addressed): I = Internet; TM = Mobile Phone; JL = Online Gaming / Video Games. Treatment: CBT = Cognitive Behavioral Therapy; AI = Artistic Intervention; EI = Educational Intervention; PE = Physical Exercise; M = Medication; FT = Family Therapy; TC = Therapeutic Camp; PT = Parenting Training; PDT = Psychodynamic Therapy. Control group: NT = Untreated / waiting list; TP = Psychological therapy; M = Medication. Measurement instruments: YIAS = *Young Internet Addiction Scale*; SAPS = *Smartphone Addiction Proneness Scale*; CERI = *Questionnaire on Internet-Related Experiences*; OGCAS = *Online Game Cognitive Addiction Scale*; CSAS = *Computerspielabhängigkeitsskala*; APIUS = *Adolescent Pathological Internet Use Scale*; IGD = *Internet Gaming Disorder*; GAST = *Game Addiction Screening Test*; S-Scale = *Youth Smartphone Self-report Scale*; IAS = *Internet Addiction Scale*; ISS = *German Internet Addiction Scale*; KFN-CSAS-II = *German Computer Gaming Addiction Scale*. TE (Effect Size) = Standardized mean difference. *n_t* (Experimental group size). *n_c* (Control group size). Cal. = Score on the quality scale; see Appendix 3.

Continuing with the description of the sample of primary studies, it can be observed that the interventions were conducted primarily with men. The highest percentage of women found in a study with a clinical population was 50.7% (Bong et al., 2021). It is also particularly noteworthy that no study was found that met the inclusion criteria and whose intervention objective was social media addiction. Table 5 shows the frequencies and percentages of new technologies targeted by the interventions.

Table 5.

Frequency of new technologies targeted by interventions

<u>New technology</u>	<u>Frequency</u>	<u>Percentage</u>
Internet	7	38.89
Video games	9	50
Cell phone	2	11.11
Social media	0	0
<u>Total</u>	<u>18</u>	<u>100</u>

Regarding the types of interventions used, their frequency and percentage are shown in Table 6. Cognitive behavioral therapy is the most common type of treatment found in the studies that met the inclusion criteria (Agbaria, 2022; Bong et al., 2021; Hong et al., 2020; S. M. Kim et al., 2012; Li & Wang, 2013; Torres-Rodríguez, Griffiths, Carbonell, et al., 2018), although in half of these cases it was used as a supplement to other types of therapy (Bong et al., 2021; Hong et al., 2020; S. M. Kim et al., 2012). In two of the aforementioned cases, the control group was also treated with cognitive-behavioral therapy, so the intervention being tested was actually the complementary one (Bong et al., 2021; Hong et al., 2020).

The label “Artistic Intervention” has been used to group two studies that employed some form of intervention based on artistic expression as a therapeutic method. Bong et al. (2021) used a music therapy intervention as a complement to therapy

cognitive-behavioral therapy, seeking to test the hypothesis that the combined effect of the two therapies would be greater than that of cognitive-behavioral therapy alone. On the other hand,

P. W. Kim et al. (2013) developed an intervention based on literary expression—both oral and written—to reduce addiction to a very specific online video game classified as an MMORPG (*Massive Multiplayer Online Role-Playing Game*).

Studies in which the intervention took place in a school setting have been grouped under the label “Educational Intervention” (del-Rey-Alamillo et al., 2012; Lindenberg et al., 2022; Uysal & Balci, 2018; Walther et al., 2014). The interventions primarily consisted of multidisciplinary approaches, with a focus on psychoeducation and the promotion of healthy usage habits. The intervention that stood out most from the others was the one developed by Uysal & Balci (2018), as it targeted a population with strong suspicions of internet addiction.

The study that combined cognitive-behavioral therapy with medication (S. M. Kim et al., 2012) used bupropion as the prescribed medication. It is worth noting that the sample in this study was also diagnosed with attention-deficit/hyperactivity disorder (ADHD). This medication is a selective catecholamine reuptake inhibitor (SSRI) frequently used in the treatment of ADHD in adolescents, major depression in adults, and smoking cessation as an adjunct to psychological therapy (Spanish Association of Pediatrics, 2020). The study, which employed a psychodynamic intervention (Shin & Jang, 2016), used Jungian sandplay therapy in a group setting.

Table 6.*Frequency of Treatment Types*

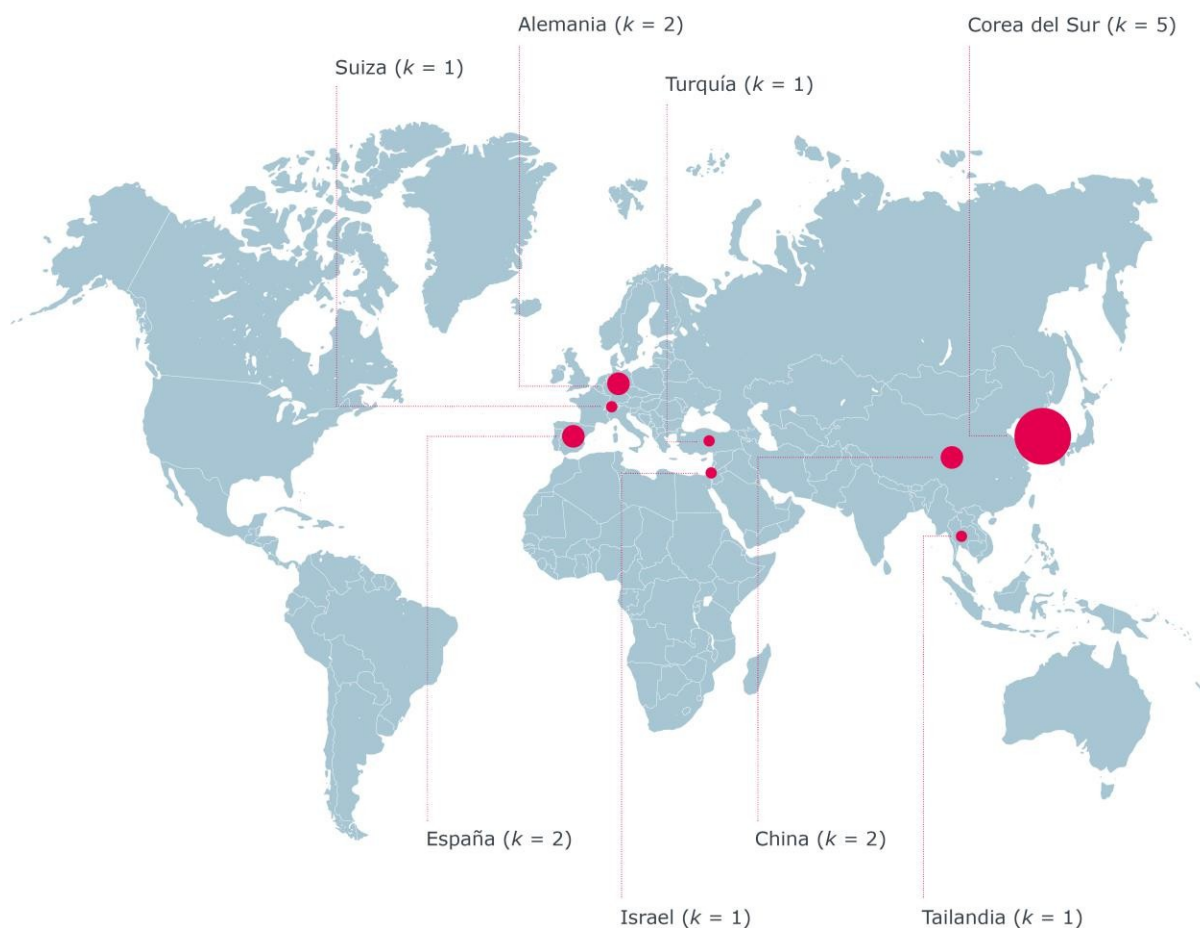
Type of intervention	Frequency	Percentage
Cognitive-behavioral therapy	6	31.58
Educational intervention	4	21.06
Family therapy	2	10.53
Art Therapy	2	10.53
Psychodynamic therapy	1	5.26
Therapeutic Camp	1	5.26
Parenting Training	1	5.26
Physical exercise	1	5.26
Medication	1	5.26
Total	19	100

Regarding the measurement instruments, the most widely used ($k = 5$) is the *Young Internet Addiction Scale* (Young, 1998), which assesses the characteristics and behaviors associated with compulsive internet use by measuring compulsivity, escapism, and dependence. The response format is a 5-point Likert scale (1–5) on which respondents indicate their level of agreement with the described items. The theoretical conceptualization of the test is that internet addiction is an impulse control disorder, with the term “internet” used to encompass all types of online activity. This is why the test has been used to measure both internet addiction (in a broad sense) and online video game addiction.

Figure 5 provides a graphical representation of the geographic distribution of the studies that met the inclusion criteria for this meta-analysis. Notable features include the large number of studies from Asia—particularly South Korea—and the complete absence of studies from the Americas. Within Western culture, a greater number of publications do appear in Europe.

Figure 5.

Geographic location of the studies included in the meta-analysis



Note. Source: Author's own work

3.2. Combined estimate of effect size

Focusing on the meta-analytic comparisons themselves, we first calculated the combined estimate of the mean effect size of the interventions included in the analysis. The point estimate of the mean effect size ($d_{\bullet}$) of treatments for addiction to new technologies is 0.725 (95% CI between 0.381 and 1.069), which is statistically significant ($z_{d_{\bullet}} = 4.129$; $p < 0.001$). The magnitude of this effect size is medium to large according to the standards proposed by Cohen (1992). The between-study variance (τ^2), obtained using restricted maximum likelihood, is equal to 0.389.

A heterogeneity test (Q) was conducted to assess the homogeneity of the estimates—that is, to determine whether the variance exceeds that expected by chance—concluding that heterogeneity does exist in the estimates of this meta-analysis ($Q_{14} = 146.79$; $p < 0.001$). By transforming the Q statistic, we obtain I^2 , which expresses the percentage of variability that exceeds what would be expected by chance. In this meta-analysis, the value of I^2 is 91.92%, which indicates high heterogeneity by standard criteria (Huedo-Medina et al., 2006).

The contrast power has been calculated for the classic effect size cutoff points for standardized mean differences proposed by Cohen (1992). For a small effect size ($\delta = 0.2$), the power is 0.307; for a medium effect size ($\delta = 0.5$), the power is 0.886; and for a large effect size ($\delta = 0.8$), the power is 0.998. If one wishes to determine how many studies would be necessary to achieve an adequate power value for all the aforementioned effect sizes, one can estimate the number of additional studies similar to those already included in the meta-analysis (Arranz-Orlandi & Botella, 2022). To achieve a power value greater than 0.80, it is estimated that 56 additional studies would be necessary.

3.2.1. Type of Population

As mentioned in Section 3.1, there are three effect sizes that refer to preventive interventions in non-clinical populations. A moderation analysis was conducted using a weighted analogous ANOVA based on a mixed-effects model to assess whether there are significant differences between the effect sizes of these interventions compared to therapeutic interventions in clinical or subclinical populations.

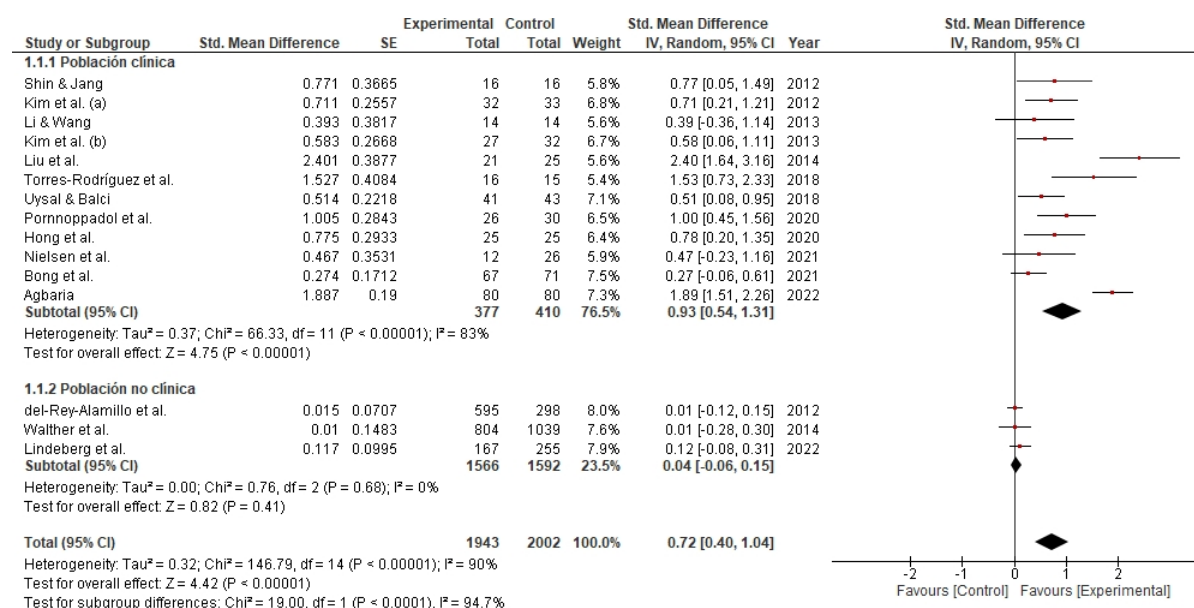
A significant difference was found between the means of the effect sizes for the two groups ($Q_B = 7.196$; $p = 0.007$). Specifically, the mean effect size

for interventions targeting the clinical population ($d_1 = 0.920$, 95% CI between 0.603 and 1.238) is greater than that for interventions targeting the non-clinical population ($d_2 = 0.048$, 95% CI between -0.505 and 0.600). An observed power of 0.99 was found for this comparison.

Additionally, it can be observed that in the group of interventions targeting clinical populations, there is still a high percentage of variance not explained by chance ($I^2_{cli} = 63\%$), whereas in the group of preventive interventions, no such variance is found ($I^2_{ncli} = 0\%$). These data suggest that in the second group there are no additional moderating variables, whereas in the first group there may still be other variables that explain the heterogeneity.

Figure 6.

Forest plot of the moderation contrast for the variable “population type”



3.3. Analysis of moderator variables

3.3.1. Gender

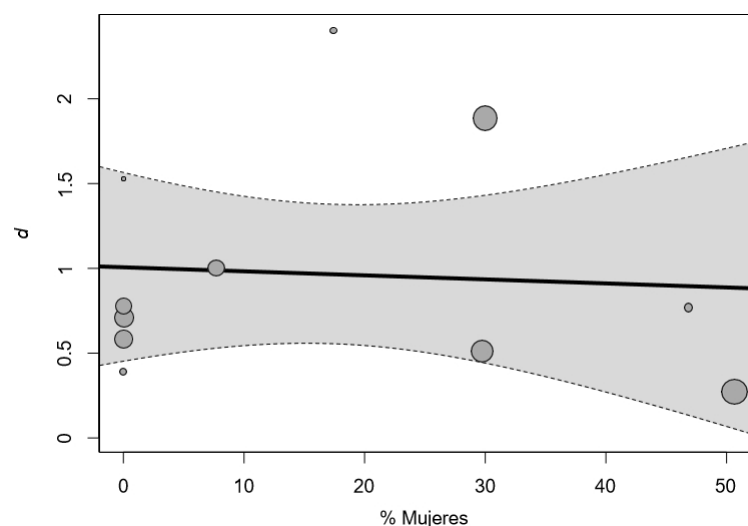
The moderating effect of sex is routinely tested in most meta-analytic studies. Since individual-level data are not available, sex is typically coded

as the percentage of women in each study. This coding results in a range from 0 to 100, quantifying a variable that is a priori categorical. The meta-regression analysis of this variable was not significant ($Q_B = 0.048$; $p = 0.827$), with a slope of $\beta = -0.002$, which is nearly zero.

It is important to note that the statistical power of these tests, in which an originally dichotomous variable is quantified, tends to be very low, especially when there is range restriction (Botella et al., 2022). Range restriction refers to the presence of numerous samples with average percentages, to the detriment of “pure” samples—that is, samples consisting of only one level of the variable. In this case, only “pure” samples consisting of men were found. This range restriction can result in very low power; therefore, it cannot be asserted that a moderating effect of sex does not exist, only that it has not been detected.

Figure 7.

Meta-regression plot for the moderator variable “sex”



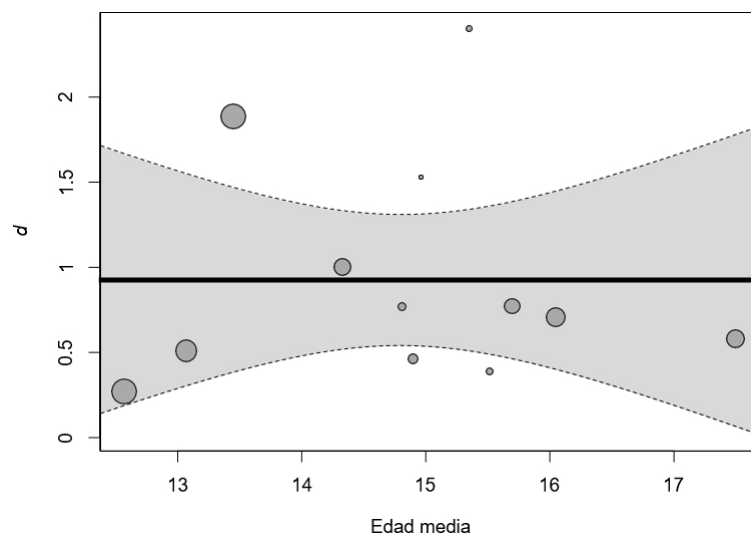
Note: The size of the circles corresponds to the weight of the studies (inverse of the variance)

3.3.2. Age

As with sex, age is a potential moderator routinely assessed in meta-analyses. Since individual data are not available, it is coded as the mean age of each study. No significant effect of this variable was found in the meta-regression analysis ($Q_B = 0.0001$, $p = 0.994$), with a slope of $\beta = -0.001$, which is nearly zero.

Figure 8.

Meta-regression plot for the moderator variable age



Note: The size of the circles corresponds to the weight of the studies (inverse of the variance)

3.3.3. Type of control group

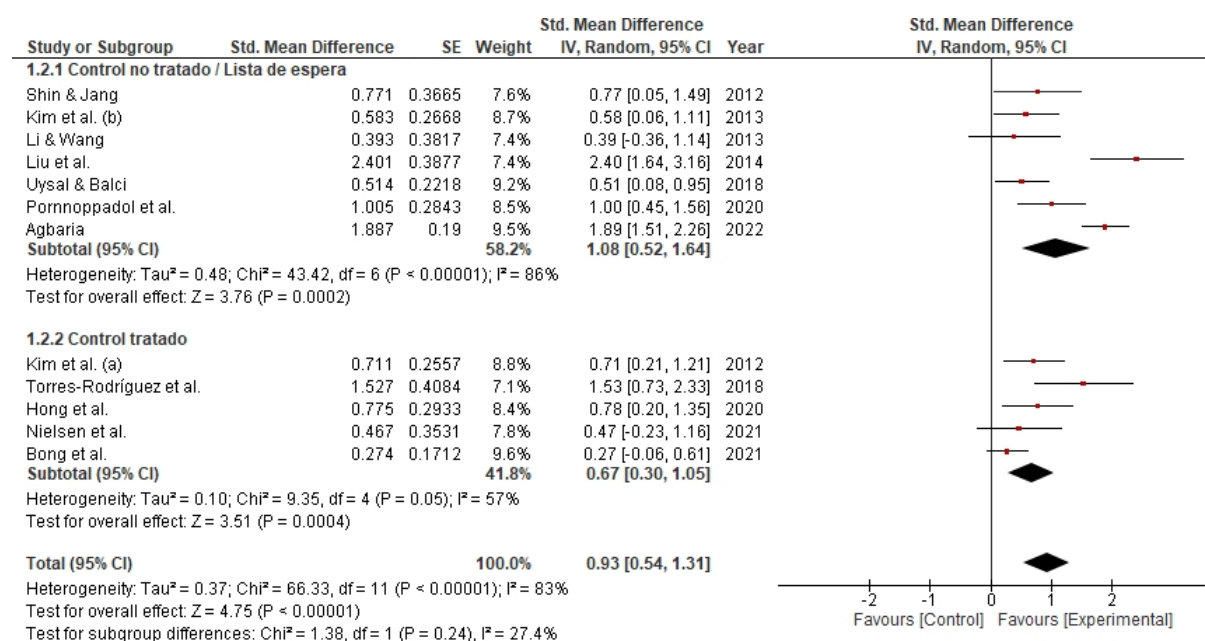
Given the small number of comparisons found in the search process, it has not been possible to directly compare the effectiveness of each intervention relative to the others. However, since several studies with treated control groups were found ($k = 5$), we decided to assess whether the type of control group has a moderating effect on effect size values. For reasons of statistical power, we grouped all

samples with an intervened control group, whether with medication ($k = 1$) or psychological therapy ($k = 4$).

No significant effect of the control group on effect sizes was found ($Q_B = 0.997$; $p = 0.318$). However, it can be observed that the mean effect size for the group whose controls had not been treated is larger ($d_1 = 1.076$, 95% CI between 0.614 and 1.538) than that of the group with treated controls ($d_2 = 0.712$, 95% CI between 0.167 and 1.257), but since their confidence intervals overlap, we cannot conclude that there are differences. An observed statistical power of 0.66 was found for this comparison. This power value is low; therefore, even if no moderating effect of the control group type is found, we cannot conclude that it does not exist.

Figure 9.

Forest plot of the moderation contrast for the “control group type” variable



3.3.4. Type of new technology targeted

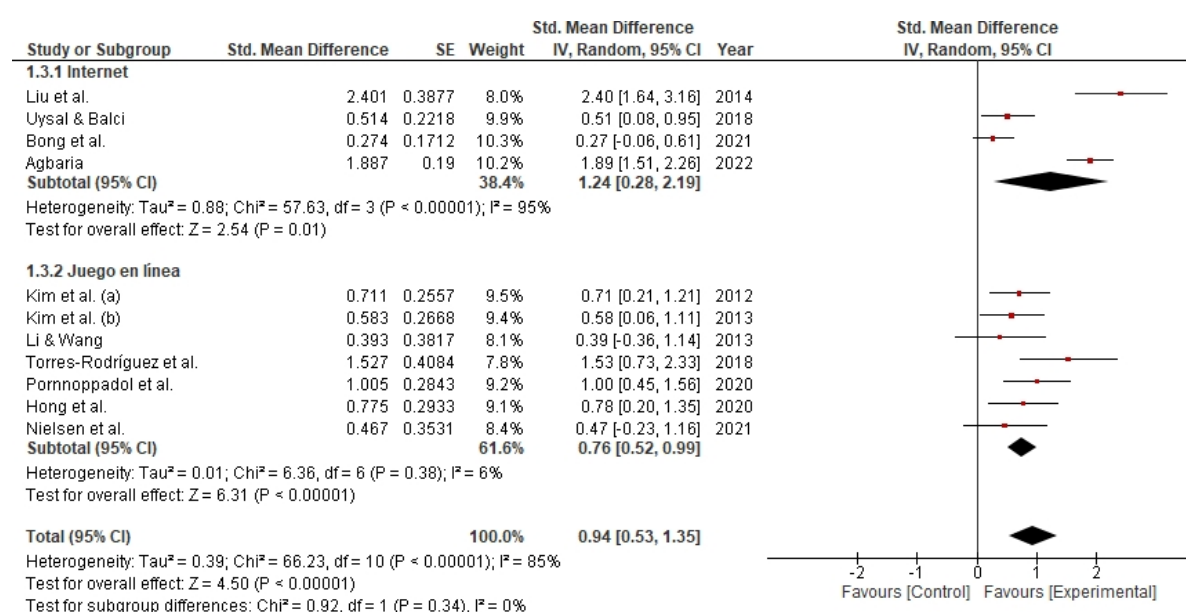
We found only studies that addressed internet addiction in a general sense, online gaming, and two studies that addressed mobile phone addiction (one

of which also addresses internet addiction (see Table 4). No studies were found that intervened on social media addiction. Given the characteristics of the data, it was decided to exclude the mobile phone addiction treatment from this contrast for reasons of statistical power. Therefore, the effect of interventions for internet gaming disorder will be compared with those for internet addiction in general.

No significant effect of the type of new technology on effect sizes was found ($Q_B = 1.099$; $p = 0.295$). The point effect size for treatments targeting internet addiction ($d_1 = 1.202$, 95% CI between 0.573 and 1.832) is estimated to be larger than that for treatments targeting internet gaming disorder ($d_2 = 0.772$, 95% CI between 0.273 and 1.273), but there is an overlap in the confidence intervals. An observed power of 0.63 was found. As with the “control group” moderator variable, such a low power implies a high probability of committing a Type II error—that is, accepting the null hypothesis (absence of differences) when it is false and differences actually exist at the real or parametric level.

Figure 10.

Forest plot of the moderation contrast for the “type of new technology” variable



3.3.5. *Type of Intervention*

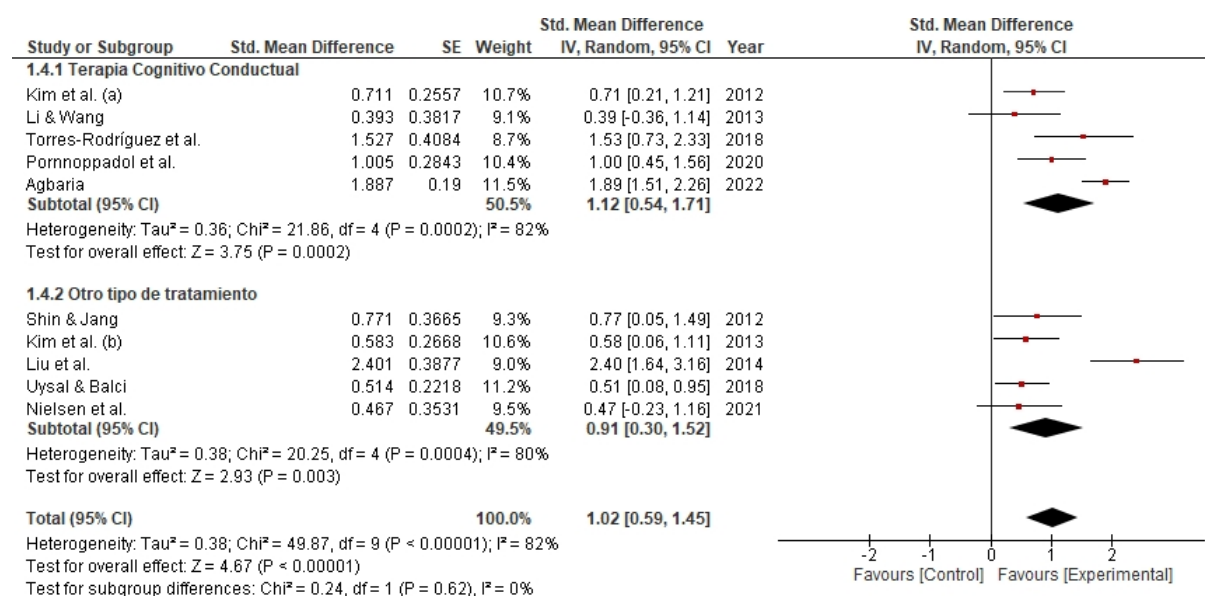
There is a vast body of literature advocating cognitive-behavioral therapy as an effective treatment for non-substance addictions in general, and for addiction to new technologies in particular. For this moderation contrast, we decided to compare studies that employed this type of intervention² with those using other therapies. This decision to group all other therapies together was made for reasons of statistical power, as there are only a limited number of studies that employ other types of therapy. Studies in which the control groups received cognitive-behavioral therapy and experimental groups consisted of this type of therapy combined with another were excluded from this analysis (see Table 4).

No significant effect of treatment type on effect sizes was found ($Q_B = 0.2561$; $p = 0.613$). A larger point estimate of the effect size is estimated for cognitive-behavioral therapy ($d_1 = 1.125$, 95% CI between 0.538 and 1.711) than for the other treatments ($d_2 = 0.910$, 95% CI between 0.317 and 1.502), but there is extensive overlap between the confidence intervals. An observed power of 0.20 was found. As with the two previous potential moderator variables, such a low power implies a high probability of committing a Type II error, specifically 0.80. Such a low power completely precludes testing the hypothesis that cognitive behavioral therapy is the most effective.

² In addition to the studies labeled “CBT” in Table 1, data from Pornnoppadol et al. (2020) have also been included, since the therapeutic camp was centered on cognitive behavioral therapy.

Figure 11.

Forest plot of the moderation analysis for the variable “type of new technology”



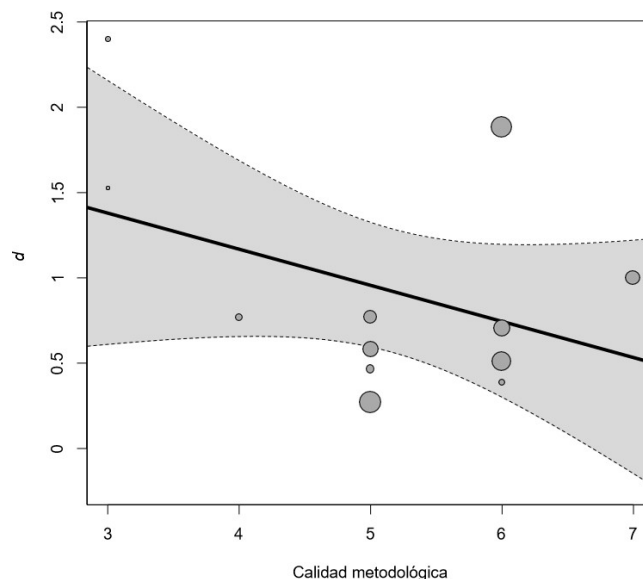
3.3.6. Methodological quality of the studies

The methodological quality of the primary studies is an extrinsic moderating variable, but no less important for that. There is no consensus on how to assess it (Botella & Sánchez-Meca, 2015), so a wide variety of instruments are used. For this meta-analysis, we selected an instrument designed for empirical studies examining the effectiveness of psychological or pharmacological treatments (López-López et al., 2013), with a score range from 0 to 10. The results of the evaluation for each study are provided in Appendix 3.

No significant effect of the methodological quality of the studies on effect sizes was found ($Q_B = 0.1723$, $p = 0.189$), with a slope of $\beta = -0.211$. Although the meta-regression analysis was not significant, it is noteworthy that two of the three studies with the largest effect sizes have the lowest quality scores.

Figure 12.

Meta-regression plot for the moderator variable methodological quality



Note: The size of the circles corresponds to the weight of the studies (inverse of the variance)

3.4. Publication bias

To assess potential publication bias in this meta-analysis, safety numbers were first calculated following the procedures of Rosenthal (1979), Orwin (1983), and Rosenberg (2005), with the results shown in Table 7. The cutoff value used for comparison was calculated using Rosenthal's (1979) original proposal. Since the safety numbers exceed the cutoff value, if we follow this method, this meta-analysis appears robust to potential publication bias.

Table 7.

Safety figures from the meta-analysis

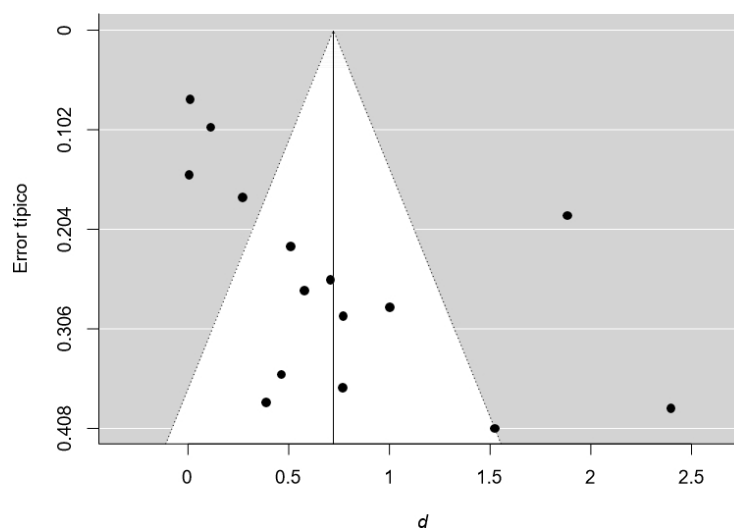
Method	NS	Crit.
Rosenthal	602	85
Orwin	214	85
Rosenberg	192	85

Note: NS = Safety number. Crit. = Criterion value. The Rosenthal and Rosenberg methods were calculated using $\alpha = 0.05$. The Orwin method was calculated using a target effect size of 0.05.

Since the safety number technique is beginning to be considered obsolete, and more sophisticated techniques exist, an analysis of asymmetry was performed using a *funnel plot* of the studies included in the meta-analysis (Figure 13). A numerical solution for this technique was calculated using Kendall's tau correlation coefficient (Begg & Mazumdar, 1994), and the result is significant ($\tau_{\text{Kendall}} = 0.467$; $p = 0.016$), and using Egger's regression test (Egger et al., 1997), which also yielded a significant result ($z = 2.376$; $p = 0.018$; $\beta = -0.128$). These results indicate the presence of asymmetry in the *funnel plot*. If the *trim-and-fill* technique (Duval & Tweedie, 2000a, 2000b) is applied to correct for this effect, no outliers are detected in the tail of the effect (right tail). Therefore, the asymmetry is due to the tail of the non-effect, a phenomenon that is the opposite of publication bias.

Figure 13.

Funnel plot of the meta-analysis



If the three studies whose intervention objective is prevention in a non-clinical population (del-Rey-Alamillo et al., 2012; Lindenberg et al., 2022; Walther et al., 2014) are considered a separate population of studies, they could be excluded from the analysis.

All analyses of publication bias are then repeated, but excluding the three studies mentioned above. The robustness statistics (Table 8) continue to indicate robustness against extreme publication bias. The new *funnel plot* (Figure 14) indicates, based on Kendall's tau correlation coefficient ($\tau_{\text{Kendall}} = 0.273$; $p = 0.250$) and Egger's regression test ($z = 0.750$; $p = 0.453$; $\beta = 0.387$), an absence of asymmetry and, therefore, of publication bias.

Table 8.

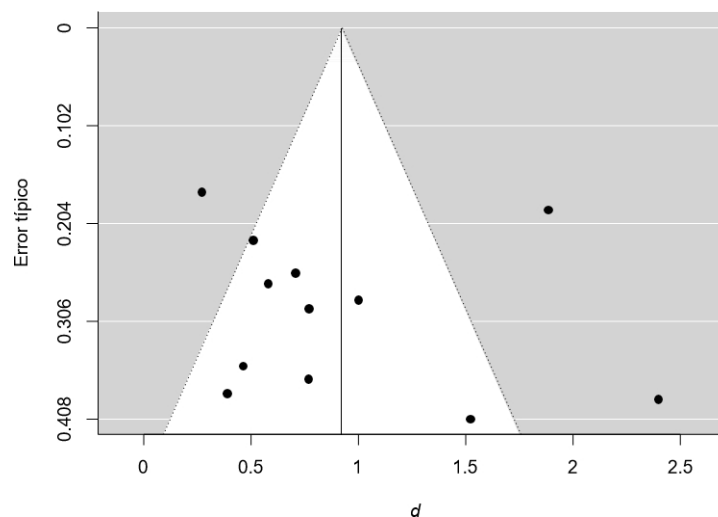
Safety data from the meta-analysis (excluding studies with non-clinical populations)

Method	NS	Crit.
Rosenthal	562	70
Orwin	215	70
Rosenberg	400	70

Note: NS = Security number. Crit. = Criterion value. The Rosenthal and Rosenberg methods were calculated using $\alpha = 0.05$. The Orwin method was calculated using a target effect size of 0.05.

Figure 14.

Funnel plot of the meta-analysis (excluding studies with non-clinical populations)



4. Discussion

4.1. The Concept of Addiction to New Technologies

Addiction to new technologies is a mental health problem that is difficult to conceptualize for various reasons (Ryding & Kaye, 2018). First, unlike other non-substance addictions such as gambling, new technologies are part of the daily lives of virtually every human being. They are an essential tool in the lives of most people, who use them either as the central focus of their profession or communication, or as a form of leisure. In this context, it is very difficult to establish clear boundaries and criteria to determine when extensive use of these technologies should be considered an addiction (Labrador & Villadangos, 2009). An example of this is the professional video game player, whether in the competitive sphere (*esports*) or the recreational sphere (*streamer* or *influencer*). For these individuals, their livelihood comes from earning money by playing video games, but they frequently report mental health problems stemming from this professional activity (Moreno, 2020).

Another major problem with addiction to new technologies is their rapidly changing nature. While the manual by Echeburúa Odriozola et al. (2009) included television as a subtype of addiction to new technologies, the manual by Carbonell (2014a), published just five years later, omits this subtype and introduces social media. Furthermore, the characteristics of each specific subtype can also change rapidly. For example, the most popular social media platforms frequently vary with the emergence of new apps, the most recent example being *TikTok* in 2016. Differences in how each app is used and its target audience complicate the design of specific interventions for addiction (Carbonell, 2020). This ever-changing nature of new technologies contrasts with the much slower scientific research process. The potential

result is a lag in academic research relative to the needs of clinical practice.

Nor is there a consensus on which types of new technologies might lead to an addiction disorder in adolescents that warrants scientific attention. Internet addiction in a general sense has not been included in diagnostic manuals because the internet is merely the medium through which potential sources of addiction—such as video games or social media—are accessed (Carbonell et al., 2012). This could partly explain the wide variability found in this meta-analysis regarding interventions for internet addiction. It is possible that the concept of internet addiction is not specific enough. Another type of new technology that has been criticized in this regard is the cell phone. Like the internet, the cell phone is seen as the medium that enables problematic use, but it is not the problem in and of itself (Carbonell, 2020). The analogy with other substance addictions would be to equate the phone with the bottle (in alcohol dependence) or the cigarette (in nicotine dependence) (Panova & Carbonell, 2018). This hypothesis is highly plausible, but further research is still needed to determine whether the cell phone possesses specific characteristics that constitute a potential source of addiction (independent of applications such as social media).

Regarding consensus, one of the criticisms leveled at the DSM-5's proposal for Internet Gaming Disorder (APA, 2014) is the exclusion of *non-online* or non-multiplayer video games (Carbonell, 2020). This consideration has, however, been taken into account in the new version of the International Classification of Diseases (ICD-11; World Health Organization, 2022). This manual has included Video Game Disorder, whether online or offline. As mentioned, there is no international consensus, although some research suggests that *offline* games have a low addictive potential (Estallo et al., 2001). Other

criticisms of the classification of these disorders is that these manuals are primarily aimed at adults, and the symptoms listed may not be appropriate for the adolescent population (Carbonell, 2020), which is the main at-risk group for this condition.

A highly controversial potential symptom of addiction is the amount of time spent using new technology. Various authors consider usage time to be a predictor or symptom of addiction to new technologies (Baggio et al., 2016; Severo et al., 2020; Tejeiro et al., 2012; Wu et al., 2013), such that the more time spent on new technologies, the greater the likelihood of developing an addiction. However, other studies indicate that the amount of time spent on new technology is not related to addiction (Charlton & Danforth, 2007, 2010; Király et al., 2017; Wood & Griffiths, 2007). Both diagnostic manuals conclude that there is no relationship between addiction and time spent, thus excluding gaming time from the symptoms of video game disorder (ICD-11) and internet gaming disorder (DSM-5).

Finally, the ubiquity of new technologies means they have a significant impact in the media, often with negative and sometimes erroneous connotations (Estallo, 2009). This situation has the potential to trigger public alarm, leading to excessive concern among parents and guardians regarding their children's use of new technologies (Labrador & Villadangos, 2009), with families being the primary source of requests for psychological care for adolescents (Carbonell, 2020). As mentioned in section 1.2.1, various measures have been taken at the policy level to try to curb or mitigate addiction among minors, especially in Asian countries, where the prevalence of addiction is much higher (Labrador & Villadangos, 2009). However, these restrictive strategies do not appear to have been effective (J. Choi et al., 2018) and have even been abolished in South Korea. Another source of

confusion is the colloquial terminology used by users and the press to refer to new technologies, particularly video games (Estallo, 2009). It is common for the quality of a game to be evaluated based on its “addictive” or “hooking” potential, referring to the high level of motivation the game can elicit.

In summary, addiction to new technologies is a mental health issue that has been a concern for both society and the academic world. However, more than 80 years after Gordon Allport expressed concern about a possible addiction to radio in 1935 (Carbonell, 2014b), there is no consensus on the concept of addiction to new technologies, and further research is still needed (Geisel et al., 2021). The most significant development—which has spurred research on this issue, specifically regarding video games—is the proposal to classify it as a disorder in the DSM-5 and subsequently in the ICD-11. This presents new challenges and opportunities for the detection, diagnosis, and intervention (Carbonell, 2020) regarding addiction to this new technology and to other, more recently emerged forms, such as social media.

4.2. Effectiveness of Interventions for Addiction to New Technologies

The objective of this study was to evaluate the effectiveness of interventions for treating addiction to new technologies in adolescents through a meta-analytic comparison. The overall finding is that current treatments appear to be effective and significantly reduce patients’ symptoms. However, these results should be interpreted with caution for two main reasons: the limitations of the included primary studies and the limitations in the conduct of the meta-analysis.

Regarding the limitations of the selected studies, it is worth noting the small number of interventions identified. Only 12 therapeutic interventions targeting adolescents were found, and these were primarily for the treatment of

internet gaming disorder. Not enough interventions were found to compare treatments for smartphone addiction and social media addiction. The small sample size also prevented us from determining which interventions are most effective. The best method for evaluating this aspect is *network meta-analysis*, which allows for direct and indirect comparisons of each pair of treatments. However, this technique requires a larger number of primary sources and demands much stricter statistical assumptions, such as sample equivalence (Fernández-Castilla & van den Noortgate, 2022).

Given the studies identified and included in the meta-analysis, it appears very difficult to conduct a network meta-analysis on this type of intervention over an extended period of time. This is primarily due to the low methodological quality of the research. According to the quality assessment tool used, the highest score was 7 out of 10, with most studies ranging between 5 and 6. Also noteworthy is the number of studies excluded due to methodological deficiencies, such as the absence of a control group. While it is true that conducting research with the highest methodological quality is challenging in minors—since, for example, it is difficult to establish an untreated or placebo control group for ethical reasons (Torres-Rodríguez, Griffiths, Carbonell, et al., 2018)—we should not cease to demand the highest methodological rigor to properly assess which interventions are most effective; therefore, more randomized clinical trials are still needed (Han et al., 2018; Stevens et al., 2019).

Regarding the limitations of the meta-analysis, first, the studies were coded by a single person, which is not recommended. Typically, coding is performed independently by at least two researchers, and the

the degree of agreement is assessed using Cohen's kappa coefficient (Siegel & Castellan, 1988). Furthermore, debatable decisions were made when conducting the moderation analyses. For many of these analyses, it was decided to group different studies together to increase the statistical power of the test, but this decision was not guided by theoretical considerations. Despite these decisions, all moderation tests (with the exception of population type) have shown very low observed power, which prevents the drawing of solid conclusions from these tests. This fact reinforces the idea that it is necessary to continue conducting high-quality primary research that allows for more accurate conclusions to be drawn regarding the effectiveness of interventions for addiction to new technologies.

Although there are not enough studies in adolescent populations that rigorously evaluate interventions, work is underway to develop treatments for these addictions by extrapolating from other populations (adults) or other non-substance addictions (King et al., 2011). In Spain, the Psychotherapeutic Intervention Program for ICT Addiction (PIPATIC; Torres-Rodríguez & Carbonell, 2017; Torres-Rodríguez, Griffiths, & Carbonell, 2018; Torres-Rodríguez, Griffiths, Carbonell, et al., 2018) is currently being developed. The fundamental paradigm is cognitive-behavioral therapy, although it incorporates strategies from other interventions such as motivational interviewing, psychoeducation, and person-centered therapy. The program is designed for adolescents aged 12 to 18, and the clinical objectives of the treatment are to reduce addictive symptoms and improve the patient's well-being.

The intervention consists of 22 45-minute sessions conducted individually and with the family. Patients are also assigned tasks to complete at home, and a family member is asked to participate as a co-therapist. The program is divided into six modules, which are

are in turn divided into submodules. Table 9 provides a summary of the program content for each submodule. It should be noted that the program is still under development and requires further empirical support.

Table 6.

Summary of the PIPATIC intervention for new technology addiction (Torres-Rodríguez & Carbonell, 2017)

Module	Submodule	Content
Psychoeducation and Motivation (3 sessions)	Motivation	Motivational Interviewing
	Goal Setting	Goal-setting to define the problem and generate the Initiating change.
	Psychoeducation with the adolescent	Organizing daily responsibilities and routines New Technologies and Video Games: Adaptive and Maladaptive Use, Pros and Cons.
	Family psychoeducation	New technologies and video games: adaptive and maladaptive use, pros and cons.
Standard addiction intervention adapted to ICT addiction (5 sessions)		Impulse control and craving. False
	Stimulus control in addiction	sense of control and self-control ability.
	Coping strategies	Inappropriate coping responses related to addiction and low self-control.
	Cognitive restructuring	Cognitive distortions and irrational beliefs that influence control.
	Problem-solving	Identifying and resolving problematic situations.
Intrapersonal (6 sessions)	Exposure	Disappearance of <i>cravings</i> .
	Identity	Self-awareness through introspection and the different levels of identity.
	Self-esteem	Rogers' three concepts of the self.
	Emotional Intelligence	Developing self-control and emotional regulation strategies.
	Coping Skills	Promoting adaptive behaviors and self-management of anxiety.

	Problem-solving	Intrapersonal and interpersonal problem-solving strategies
Interpersonal (2 sessions)	Identity	Communication Styles
	Assertiveness	Basic Human Rights
	Response Style	Repetitive patterns of behavior, thought, and emotion.
Family (4 sessions)	Family communication	Adaptive family communication styles.
	Setting Boundaries	Family functioning and the importance of family boundaries.
	Building Bonds	Family attachment: insecure, avoidant, disorganized, and secure.
Creating a New Lifestyle (2 sessions)	Self-Observation of Improvement	Changes and skills.
	Alternative activities	Alternative leisure activities
	Relapse prevention	Simple relapse prevention therapy.

Given the rise of new technologies and adolescents' easy access to them, it is of great importance to implement large-scale preventive interventions aimed at reducing the incidence of addiction to new technologies (King et al., 2018). These interventions, which still require further research and refinement (King et al., 2018), are typically carried out within an educational setting.

A good example of such interventions is the PROTECT program (Professioneller Umgang mit technischen Medien [Professional Use of Technological Media]; Lindenberg et al., 2022), which is currently being developed in Germany. This intervention is primarily based on the cognitive-behavioral paradigm with additional components of psychoeducation and emotional regulation. The intervention consists of 16 sessions (four per week), each lasting 90 minutes, and is delivered in groups of six to 10 students. The program is divided into four modules: (1) boredom and motivational problems; (2) procrastination and performance anxiety; (3) social interaction; and (4) emotional regulation.

In Spain, the only intervention identified is the ConRed program (del-Rey-Alamillo et al., 2012), whose primary objective is the prevention of cyberbullying. However, its objectives also include “preventing the misuse of ICTs and highlighting the consequences of technological dependence” (p. 133). The theoretical framework of the intervention is based on normative social behavior theory (Rimal & Real, 2003). According to this theory, beliefs are mediated by the following normative mechanisms: (1) recognized legal norms; (2) expectations; and (3) group identity. To modify a specific type of behavior, all three mechanisms must be addressed. This program is also divided into four modules: (1) the educational community’s prior knowledge regarding the use and functioning of the internet; (2) privacy, identity, and opportunities on social media; (3) prosocial behavior, solidarity, and risks on social media; and (4) coping strategies for problems on social media.

In summary, there are numerous interventions designed to treat or prevent addiction to new technologies. These interventions are based on findings from research on other behavioral addictions, such as gambling (King et al., 2011), but lack sufficient evidence of their own, particularly among adolescents (Han et al., 2018; Stevens et al., 2019). The meta-analysis conducted in this study supports this latter point, as only a small number of studies evaluating these interventions were identified. Furthermore, many of these studies suffer from methodological limitations and shortcomings, meaning their results must be interpreted with the utmost caution. Nevertheless, it is important to note that this field of study is very active, and the inclusion of video game disorder (ICD-11) and internet gaming disorder (DSM-5) represents a significant advance in the conceptualization, detection, and intervention for addiction to new technologies.

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³ The references for the studies included in the meta-analysis are those preceded by an asterisk (*)

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6. Appendices

6.1. Appendix 1. R Code

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#####
## META-ANALYSIS CODE ## ##
M.P.G. SANITARIA UI1 THESIS ##
##          ---          ##
## Santiago Arranz Orlandi ##
#####

# Required package
# install.packages("metafor") # Run this the first time to install the
package
library(metafor)

# Read the complete data
df <- read.table("base_coded.csv", header = T, sep = ";")

## COMBINED ESTIMATION OF TE ##
df.com <- df[order(df$year),] # Studies sorted by year #
Calculation (We assume an EA model estimated using REML)
ma <- rma.uni(yi=es, vi=var, data=df.com, method="REML", weighted=T)
summary(ma)
# Power source(power.R)
source(studies.power.R)
power.ma(ES=es, var=var, data=df.com)
studies.power(ES=es, var=var, data=df.com, power=0.80)

## MODERATOR: POPULATION ##
# Calculation in
SPSS # Power
v1.goal <- (0.1620^2)/12
v2.goal <- (0.2820^2)/3
lambdab.goal <- ((0.9201-0.0477)^2)/(v1.goal + v2.goal)
pot.goal <- 1 - pchisq(q = qchisq(p=0.95, df=1), df = 1, ncp =
lambdab.goal)

# Data frame with clinical population data df.mod <-
df.com[!df.com$goal==2,]

## MODERATOR: Sex ##
df.wom <- df.mod[order(df.mod$wom),] # Studies sorted by the moderator
df.wom <- df.wom[!df.wom$wom==999,] # Remove missing cases
```

```

# Calculation (We assume a mixed model estimated using REML)
ma.wom <- rma.uni(yi=es, vi=var, data=df.wom, method="REML", weighted=T,
mods=wom)
summary(ma.wom)
# Plot
regplot(ma.wom)

## MODERATOR: Mean age ##
df.age <- df.mod[order(df.mod$age),] # Education data sorted by the
moderator # Calculation (We assume an EA model estimated using REML)
ma.age <- rma.uni(yi=es, vi=var, data=df.age, method="REML", weighted=T,
mods=age)
summary(ma.age)
# Plot
regplot(ma.age)

# MODERATOR: Type of control group
# Calculation in SPSS
# Power
v1.cont <- (0.2357^2)/7
v2.cont <- (0.2782^2)/5
lambdab.cont <- ((1.0758 - 0.7118)^2)/(v1.cont + v2.cont)
pot.cont <- 1 - pchisq(q = qchisq(p=0.95, df=1), df = 1, ncp =
lambdab.cont)

## MODERATOR: Type of new technology ## #
Calculation in SPSS
# Power
v1.tech <- (0.3213^2)/4
v2.tech <- (0.2548^2)/7
lambdab.tech <- ((1.2023 - 0.7724)^2)/(v1.tech + v2.tech)
pot.tech <- 1 - pchisq(q = qchisq(p=0.95, df=1), df = 1, ncp =
lambdab.tech)

## MODERATOR: Type of intervention ##
# Calculation in SPSS
# Power
v1.treat <- (0.2992^2)/5
v2.treat <- (0.3022^2)/5
lambdab.treat <- ((1.1248 - 0.9096)^2)/(v1.treat + v2.treat)
pot.treat <- 1 - pchisq(q = qchisq(p=0.95, df=1), df = 1, ncp =
lambdab.treat)

## MODERATOR: Methodological quality ##
df.qual <- df.mod[order(df.mod$qual),] # Studies sorted by the moderator
# Calculation (We assume an EA model estimated using REML)

```

```

ma.qual <- rma.uni(yi=es, vi=var, data=df.qual, method="REML", weighted=T,
mods=qual)
summary(ma.qual)
# Plot
regplot(ma.qual)

### PUBLICATION BIAS ###
## With all studies ## #
Safety data
fsn(yi = es, vi = var, data = df.com, type = "Rosenthal") # Rosenthal
method
fsn(yi = es, vi = var, data = df.com, type = "Orwin", target = 0.05) #
Orwin method
fsn(yi = es, vi = var, data = df.com, type = "Rosenberg") # Rosenberg
method
crit <- 5 * nrow(df.com) + 10 # Criterion value calculated using the
Rosenthal method
crit
# Funnel plot
funnel.rma(ma)
# Kendall's tau correlation coefficient ranktest(ma)
# Egger's regression
test regtest(ma)
# Trim and fill
trimfill.rma.uni(ma)

## For studies with clinical populations ##
# Safety data
fsn(yi = es, vi = var, data = df.mod, type = "Rosenthal") # Rosenthal
method
fsn(yi = es, vi = var, data = df.mod, type = "Orwin", target = 0.05) #
Orwin method
fsn(yi = es, vi = var, data = df.mod, type = "Rosenberg") # Rosenberg
method
crit <- 5 * nrow(df.mod) + 10 # Criterion value calculated using the
Rosenthal method
crit
# Funnel plot
ma.mod <- rma.uni(yi=es, vi=var, data=df.mod, method="REML", weighted=T) #
Combined estimate of the TE for the other functions
funnel.rma(ma.mod)
# Kendall's tau correlation coefficient
ranktest(ma.mod)
# Egger's regression

```



```
regtest (ma.mod)
```

```
####END####
```

6.2. Appendix 2. SPSS Syntax

```
* SYNTAX FOR THE META-ANALYSIS 'TFM M. P.G.SANITARIA UI1'.
```

```
* SANTIAGO ARRANZ ORLANDI.
```

```
* MODERATOR: Population type.
```

```
INCLUDE"F:\Users\arran\Documents\Psychology\Master's Psychology General  
Health\TFM\MetaF.sps" .
```

```
METAF ES= es /W= w /Group= goal /MODEL= REML .
```

```
* MODERATOR: Type of control group.
```

```
* Grouping of studies with a treated control  
group. RECODE cont (1=1) (ELSE=2) INTO cont_rec.
```

```
INCLUDE"F:\Users\arran\Documents\Psychology\Master's General General  
Health\Master's Thesis\MetaF.sps" .
```

```
METAF ES= es /W= w /Group= cont_rec /MODEL= REML .
```

```
* MODERATOR: Type of new technology.
```

```
* Recoding.
```

```
RECODE tech (1=1) (2=2) (13=1).
```

```
INCLUDE"F:\Users\arran\Documents\Psychology\Master's Psychology General  
Health Psychology\TFM\MetaF.sps" .
```

```
METAF ES= es /W= w /Group= tech /MODEL= REML .
```

```
* MODERATOR: Type of treatment.
```

```
* Recoding.
```

```
RECODE treat (1=1) (23=1) (16=1) (ELSE=2) INTO treat_rec.
```

```
INCLUDE"F:\Users\arran\Documents\Psychology\Master's Psychology General  
Health\TFM\MetaF.sps" .
```

```
METAF ES= es /W= w /Group= treat_rec /MODEL= REML .
```

6.3. Appendix 3. Assessment of the methodological quality of the studies included in the meta-analysis.

Item	Agbaria et al. (2022)	Bong et al. (2021)	Del-Rey- Alamillo et al. (2012)	Hong et al. (2020)	Kim et al. (2012)	Kim et al. (2013)	Li & Wang (2013)	Lindenberg et al. (2022)	Liu et al. (2015)	Nielsen et al. (2021)	Pornnoppadol et al. (2020)	Shin & Jang (2016)	Torres- Rodríguez et al. (2018)	Uysal & Balci (2018)	Walther et al. (2014)
1. Random assignment to groups?	X	X	-	X	X	X	X	X	-	X	-	-	-	X	X
2. Was it verified that the groups were matched on relevant variables in the pretest?	-	X	-	X	X	-	X	X	-	X	X	X	-	X	X
3. Were pretest measures of the dependent variable(s) included?	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
4. Was a psychological placebo control group included?	X	-	-	-	-	X	X	X	-	-	X	-	-	-	-
5. Was a pharmacological placebo control group included?	-	-	-	-	X	-	-	-	-	-	-	-	-	-	-
6. Was the evaluator masked?	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
7. Were validated assessment tools used?	X	X	X	X	X	-	X	-	X	X	X	X	X	X	X
8. Large sample size?	X	X	X	-	-	X	-	X	-	-	X	-	-	X	X
9. Post-test mortality rate equal to or less than 10%?	X	-	X	X	X	X	X	X	X	X	X	X	X	X	X
10. Were intention-to-treat analyses intent-to-treat analyses?	-	-	-	-	-	-	-	X	-	-	X	-	-	-	-
Sum	6	5	4	5	6	5	6	7	3	5	7	4	3	6	6

Note: To determine the sample size, the distribution of sample sizes in the MA studies is dichotomized based on the median, and sample sizes greater than that value are assigned the “high” category. “X” indicates a “Yes” response on the *checklist*, and “-” indicates a “No” response.